# I-ROBOT TO WE-ROBOT:

# EXPLORING THE EFFECTS OF

# TEAM STRUCTURE ON TEAM

# DYNAMICS, DECISION MAKING, AND PERFORMANCE,

# WHEN WORKING WITH A REMOTE ROBOT

By

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*This thesis represents my own work in accordance with University regulations.*
Alec Jacob Ranzato, 2016

## Abstract

In this study I gave teams of five a team structure, condition or hierarchy, and tasked them with controlling a remote robot through an experimental space with the goal of maximizing exploration and exploitation. They were given 90 minutes, and within that 90 minutes had a series of (max) 10-minute windows to send a series of commands to their robot in bulk consisting of movement and pictures, which were used to help in following planning sessions. Ultimately, three groups emerged, tightly coupled hierarchical and consensus ones, and loosely coupled versions of both. Loosely coupled teams proved to be best suited for the task as they could maintain brief social order centered around their robot teammate where their team structure provided little. They could then neutralize the advantages of their given structure and adopt the advantages of the other.

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## 1. Introduction

At a minimum distance of 56.4 million kilometers from Earth, and a maximum of 401 million, Mars is hardly a suitable place for sociology. However, on that (so far) lifeless planet are two awkward, golf cart sized, scientific tool wielding, representations of life and a number of complex social structures here on Earth. One is still roaming the surface far past its due date, searching for answers. “Roam” and “search” are apt words to describe the actions of the Mars Exploration Rovers *Spirit* and *Opportunity (*although *Spirit* is no longer active) because it gives a sense of agency to them; a personality that it has been instilled with over time, not just by the teams behind them, but as a piece of cultural history, as indicated by the comic in figure 1.

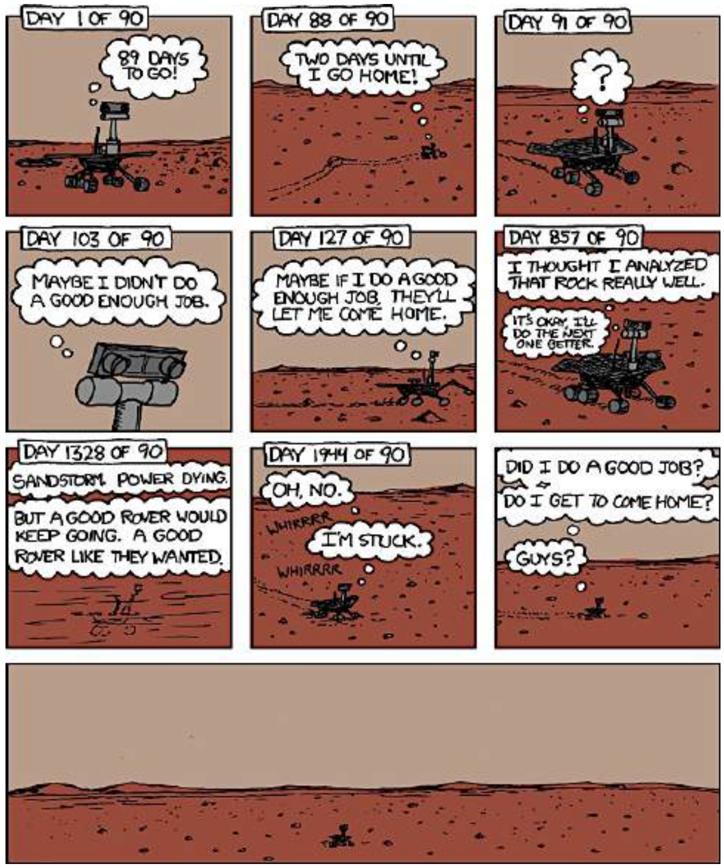


**Figure 1: XKCD Comic (https://xkcd.com/695/)**

It is true though, the Mars Rover has no artificial intelligence, no agency of its own, but it is undoubtedly an invaluable and beloved member of the Mars exploration team. The idea of a team is in fact the central theme of this thesis. Not only is the Mars Rover a product of advanced science and good teamwork, but a source of this very same teamwork.

How has the Mars Exploration Rover Mission been so successful? In Janet Vertesi's book *Seeing Like a Rover* (2015), her in depth ethnography of the Mars exploration team reveals a complex system consisting of a team uniquely bonded via "embodying" the Mars Rover, facilitated by collectivist culture and a flattened, consensus-based power structure, mediated and validated by the images produced by the rover and worked upon by the team.

It is clear that this collaborative unit stands alone among other NASA projects, which usually function through (or are inundated by) stricter hierarchies and bureaucracies. In fact, given the scale, stakes, and cost of the Mars Exploration Rover Mission, it would be impossible to even use these rovers as testing focal points studying different kinds of hierarchical arrangements, and how they affect the productivity and teamwork dynamics of teams that work with these robots.

This is where my research aims to fill the gaps. There is a sizeable body of current literature in sociology, psychology, and engineering, describing what teamwork is, what makes teamwork work (Salas, Sims, and Burke 2005), and even the effects of team structure or communication structure on team productivity (Gao, Cummings, and Bertuccelli 2012; Urban et al. 1995). However, very little of this research involves larger teams, and even less address teams that are working on a collaborative task where "all members are interdependent because of their need to collaborate through all aspects of the team task" (Salas et al. 2005:572). Finally, and perhaps most salient, is that few, if any, study teamwork based around control of a remote robotic agent. The studies that do (Gao et al. 2012), do not involve large team sizes, utilize computer simulation, and do not focus on giving subjects a wider array of points to make decisions about. Additionally in these studies, the robotic element is underplayed in the wake of more generalized research questions oriented towards team productivity in any task.

What is unique about the Mars Rover teams is that not only are the teams dependent on the rover itself to do science and must take into consideration the restrictions and limitations of this teammate, but also on any given day, the number of decisions that can be made is considerable. Although there are looming goals of exploration and exploitation embedded into the rover—every tool and design decision is made toward this end—sub-teams of specialized scientists have their own diverse set of goals that can aim the rover in numerous possible directions before a final set of commands is completed.

It is important to understand the difference between the flat, consensus-based model of the Rover teams and deeper hierarchies demonstrated in numerous other areas of science, design, and engineering. Consensus-based teams not only require that their members all agree on a course of action, but each team member generally has the knowledge and resources to complete a number, if not all of the diverse tasks associated with his or her team. This is opposed to deeper hierarchies, or sometimes more specialized “product teams” that have divisional skill and resources (Urban et al. 1995). In many studies however, team structure is intimately related to the division of work into subtasks, and thus for my research it may be easier to use the term communication structure, or restrictions on communication and the directional flow of information, when referring to collaborative work conducted by the Mars Rover teams and this study.

In fact, this collaborative work, coming to unanimous decision about where to send their robotic teammate, comprise most of the day for those situated millions of miles behind the Mars Rover. As a team they analyze and work over images produced by the rover and have become intimately aware of the Robot's "physiology." This adoption of the rover's physiology is apparently both facilitated by, and strengthens the consensus-based team planning.

Again however, without a similar project to compare to, we can't yet conclude that the success of the rover, or perhaps the success of a collaborative exploratory task, is dependent on that flat hierarchy and it begs the question of what would change should the communication of the team behind the robot change.

My study intends to explore what aspects of teamwork and success change under differing communication or team structure for teams that work with robots. How are aspects such as leadership and the "shared mental model" affected by team structure? How are patterns of communication and language affected by team structure? And finally, how will a teams' relation to their remote, robotic counterpart change when their team structure changes at home, and does this relation saliently affect the basic team dynamics expressed above?

Simulating the remote exploratory task is a good starting point for a collaborative task that intimately involves a robot. However, large advanced teams will continue to send robots where people cannot go and designing the best possible teams for these projects will be paramount to their success. These fields include earth-bound research, search and rescue, manufacturing, military operations, and medical procedures. Thus my research is a step to understanding the ideas of teamwork and team productivity in a world increasingly experiencing and explored by humans and robots working collaboratively.

## 2. Literature Review

### 2.1 The Mars Rover

"I argue that visualization practices, with their associated gestures and embodied narratives, are not only enrolled in understanding the object or instrument of analysis, but are also *organizational practices* that produce and maintain the social order of the laboratory" (Vertesi 2012:396). This quote in many ways sums up the complex success that has become the Mars Exploration Rover mission. As Vertesi explains, the teams' need to embody the rovers *Spirit* and *Opportunity* in order to successfully produce the images that are the currency between team members has also become a right of passage onto these teams themselves. The rover itself has been a site for the development of good teamwork.

On the ground, the Rover team is organized as a flat, consensus-based, team that, in the beginning, did much of its work in a very collocated fashion (Tollinger et al. 2004). This structure is reflected throughout their team processes. Collaborative planning sessions are one such ritual. Through repeat sessions over time, the group has become more efficient or expert in these proceedings reducing daily “sol planning sessions” from 8.5 to just 2 hours (Tollinger, Schunn, and Vera 2006). Team members have been able to effectively take images that, as Janet Vertesi suggests, are paramount to team coordination, communication, and solidarity, and express a wide variety of goals and scientific values to their team, in order to convey a set of ideas towards a decision. Further reflecting the centrality of the team structure, Vertesi points patterns of speech, to both human and robot teammates that convey solidarity in consensus. Referring to the robot as “we” and ending each planning session with the notable call and response pair of “are you happy” “situate the entire team behind the rover’s eyes” (Vertesi 2012:405), and make it clear that the success of this missions is married to the successful practice of the teams given structure.

Thus we can see a complex set of team dynamics operating behind the Mars Rover but before we can simulate similar conditions while altering team structure, it important to further examine the literature around team dynamics and teamwork (without a robotic counterpart) as well as understand the contemporary context of robotic teammates in the real world. This literature begins on a large sociological scale starting at the organizational level.

**2.2 Preludes for Teamwork: Solidarity and Organizational Forms**

Within the classical canon of sociology, Durkheim and Weber both establish broad scale mechanisms and models of organizations. The prelude to teamwork and even to society begins with social solidarity and Durkheim posits two types of solidarity. The first is mechanical solidarity born from homophily among group members and similarities in roles within the group. The second is the more complicated organic solidarity, which arises from the division of labor, in which group members of differentiated roles are interdependent for the maintenance and progression of the society. “No individual is sufficient unto himself,” Durkheim explains as society requires and is validated by the division of labor (Durkheim 1984:173). These two “species” of solidarity seem contentious at the large scale at which Durkheim is constructing his theory, but at the task level, explained in the following section, both of these types are important for successful teamwork, as collaboration often requires a mix. The requisite levels of each in this mix are again highly case dependent.

Max Weber, observing what he saw to be the increasing rationalization of the social world, postulated an “ideal type” of organizational form that would arise from this progression. Thus Weber’s body of work is very concerned with the

Bureaucracy as a model type of rational organization. Ultimately the bureaucracy is highly stratified in terms of labor type, rule-based at its very core, and most relevantly, hierarchical, where "each lower office is under the control and supervision of a higher one" (Weber 1968:218). Not only is the power dynamic a vertical one but also the flow of information follows the same pattern. It should be mentioned that today Weber's model is often regarded as cumbersome and odds with the "progressive" flat models of today's largest startups, but Weber saw bureaucracies as the most rational and efficient model. However, he equally feared the consequences of rationalization as increasingly separating people from one another and what this could mean for something such as solidarity. More recent scholarship has even suggested that we have moved beyond the bureaucracy altogether, placing us in a post-bureaucratic age increasingly characterized by virtualization, hidden structural elements, and increasing flexibility around pieces of the existing hierarchy (Nohria and Berkley 1994).

History has given us examples of these theories being put to the test, demonstrating the challenges of establishing successful organization and progress, and the advantages and disadvantages of their many forms. How does a group choose to be a collective consensus like the MER teams or a centralized hierarchy like the one described by Weber? The Student Nonviolent Coordinating Committee experienced this problem throughout the mid 1960s as their group expanded, civil rights concerns grew, and students both black and white wanted to be active on both the local and more national level. Francesca Polletta explains that the struggle to maintain a flat, "participatory" democracy was wrought with difficulty as it became "too unwieldy for a group grown in size" (Polletta 2002:88). The commitment to empowering people at the local level, was challenged by those wishing for a centralized hierarchy; labeling it as inefficient, impotent, and even at odds with some "black values," where a participatory democracy was seen as "white." The other side of course said the opposite, and the organization as a whole was characterized by months of stagnation and repetitive deliberation. Even this deliberation, at this time of structure-in-flux, was characterized by low levels of coordination and "organization," as meetings of 12 hours or more concerned themselves with deciding on how to decide (Polletta 2002).

Organizational forms do not always visibly conflict either. Sometimes they more subtly interact with each other where aspects of one or the other become more conspicuous at a given time. The panels of professionals in charge of "peer-reviewing" academic paper submissions technically function with consensus-based approval process. However within this system other power structures exist that are central to moving the deliberation along (Lamont 2010). Lamont describes the alliances and deals made between panelists to vote in specific direction. Additionally she explains the culture of deference to expertise and the hierarchies of power that come with it.

Further still, beyond organizations coping with, and acting in and around the formal structures consensuses and hierarchies, some organizations have developed new forms of structure that are best suited to their specific needs. In the realm of tech startups, David Stark (2009) reveals a trend towards flatter hierarchies, or heterachies. Here, *interdependent* teams, with diverse values sets, work towards collaborative engineering projects. Stark states that the rivalries and "friction" present between these laterally organized groups, is the primary source of creative and effective solutions (Stark 2009). The collaborative nature of a heterarchy thus separates it from a traditional hierarchy, but at the same time, the foundation upon interdependency does not allow one to say, "that the heterarchical firm radically decentralized decision making" (Stark 2009:102), as one might see present in other flattened structures.

The "Burning Man" festival is another example of radical new structure, and here the formal restrictions of structure are dispensed with as much as possible. In an effort to cope with the growth of Burning Man, some dose of structure and formal organization was needed, but in many ways this conflicted with the ideals of the event itself, which held freedom and the lack of formality in high regard. Katherine K. Chen (2009) thus describes the "Do-Ocracy" in which outside of loose top-down authority, members of the “community” can and are encouraged to enact grassroots projects and “activity[s] or project[s] that addressed a ‘civic need,’” such as those concerning waste disposal or bathrooms (Chen 2009:55). Within these projects decision-making and team structure become flexible, as organizers establish the norms of each team. The “do-ocracy” creates a new formal structure that operates upon making formal structure invisible so that values and pragmatism don’t come into conflict. People are able to take action towards necessary goals without directly interacting with top down authority.

It is clear then, that organizations of a variety of sizes deal with the effects of structure everyday, and success and solidarity are fundamentally linked to their practice. I want to shrink now from the larger idea of “organization” to the smaller, more intimate idea teams, where groups are working together on a specific task or set of tasks. Success at this task level is often summarized by “good teamwork,” and if that is the case it is important to describe what teamwork is and how it too is related to structure and organizational forms.

### 2.3 Teamwork and Team Structure at the Task Level

It is important to frame the Mars Exploration Rover Mission within the larger scope of teamwork and team structure. It is clear that the success of the mission can in part be attributed to good teamwork and team dynamics, but what are the accepted features of good teamwork? Additionally how has team effectiveness been studied and measured before? Furthermore, if the Mars Rover mission is a successful experiment in team structure as well, how has team structure more generally played into team effectiveness and the development of good teamwork?

Teamwork has most effectively been broken down into a five-step interplay between leadership and the ability of team members to account for each other, and adapt to changing task goal (Salas et al. 2005). Not only do leaders have to coordinate the diverse knowledge and skills of their team, but also it is imperative that they create a culture where team members hold each other responsible and take action to make sure each member does their own task, or can instead do it themselves. Important throughout the work of Salas, Sims, and Burke, is that team effectiveness is not only related to whether the task is completed, but instead how the team interacted and developed a system for completing the given task. Imperative to utilizing the “Big 5 of Teamwork” is the development of shared mental models (MacMillan, Entin, and Serfaty 2004; Mickan and Rodger 2000; Salas et al. 2005; Urban et al. 1995) a concept used by many sociological frameworks surrounding teamwork. For Salas et al, these models facilitate a shared understanding of the goals, tasks, and coordination of the team and “create a framework that promotes common understanding and action” (Salas et al. 2005:566). The presence of effective shared mental models is often correlated to high levels of team productivity and effectiveness and some indicator of these models include low levels of superfluous communication and patterns of communication that are more predictive and less "requestive" (MacMillan et al. 2004).

Teamwork and its parts are expressed differently for different types of tasks as well. The literature is split in its focus of coordinated and collaborative tasks where the former involves team members working more independently, with different sets of resources on a division of sub tasks, while collaborative tasks involve team members that are engaged in the whole process of the completion of the entire task and its sub tasks (Salas et al. 2005). The Mars Rover, and my study inspired by it, is a collaborative task. Thus, leadership doesn't assign roles and maintain lines of strict responsibility, but instead manages the lines of responsibility between team member and dictates when roles change and team members overlap. With the softer distinctions between the roles and responsibilities of team members, collaborative tasks involve more real time feedback and communication and there is a greater expectation of error management and cross-member accountability (Salas et al. 2005). Backup behavior, "the extent to which team members help each other perform their roles" (Gao, Cummings, and Solovey 2014:441), is one such feature of teamwork that is clearly easier to express and observe in a collaborative task than a coordinated one because of the nature of teammates working more explicitly together.

Teamwork is in many ways dependent on how a team is organized around a task, and the demands of the task itself. Thus not only has the general development of teamwork found good foundation within the sociological literature of team dynamics, but also how characteristics of work structure, team structure, and communication structure play into an effective team. Much of this research has been aimed at discovering tools for creating "optimal" teams for a given scenario and, as discussed in the following section, this has become of special interest with the growing introduction of robots in various human teams.

MacMillan, Entin, and Serfaty (2004), demonstrate the centrality of task requirements and resource management when deciding upon the most effective team structures. In one set of diverse tasks, optimizing resource allocation and setting up more independent task units within a separated hierarchy, was shown to be more successful than flatter allocation. Coordination was minimized through providing units with their own set of usable resources. In a separate set of task however, this same distribution of resources proved more ineffectual than coordinated single resource teams.

The study from MacMillan et al. (2004) is most relevant for its insights into the costs and benefits of communication. Generally, when resources would allow for low levels of coordination and communication, than those teams should be more effective given the extra effort it takes to communicate, and further effort still to do it effectively. However, if resources are such that separated units cannot operate without interdependence, "functional" teams, those designed with coordinated sub-teams of limited resource variance, prove to have higher levels of coordination and more efficient communication.

From this work, my study takes particular interest in the idea of "anticipation ratio" which is a ratio of communications providing, versus requesting, information. MacMillan et al. suggest that teams that "push," or deliver information more often than request it, are more efficient. These teams eliminate the extra cost of the request-receive communication pair and likely have better shared mental models of the task and their team. It is still unclear how patterns of communication and other team dynamics demonstrated by MacMillan et al. will vary when the team is collaboratively in control of only one resource, the robotic teammate, and must coordinate how to manage it towards the completion of a diverse set of goals.

Urban, Bowers, Monday, and Morgan (2009) also examine team structure and team performance, however they aimed at how to optimize teams that cannot be restructured, and what elements would allow a given structure to succeed. While their results support prior research that nonhierarchical teams are more effective, it is suggested that it's possible to remedy communication behavior on hierarchical teams to increase effectiveness. Flat teams were found, like in MacMillan et al., to ask fewer questions and make fewer requests. If Urban et al. had applied the anticipation ratio framework it appears they would find that nonhierarchical teams again “push” more, and are more efficient. Urban et al. suggest then, that in situations where reorganizing team structure is an impossibility, hierarchical and product based teams need to better predict the needs of their teammates and be highly explicit in their requests. This would hopefully reduce the communication overhead described in MacMillan et al.

It becomes apparent that although more general research suggests that flat hierarchies are more successful, this is highly reliant on the context of the task. Adding in the concept of workload busyness and team retention (team members who work together more than once), we see that workload busyness makes the formation of team mental models for hierarchy-based team more difficult. However, for similar levels of team mental models in both flat and hierarchical teams, flat teams perform better. Given the interaction between team mental model formation and team retention, this suggests that teams that have never worked together before should be organized into hierarchical or product based teams (Singh et al. 2011). In fact this context is very important to Singh et al., who note that team building versus productivity is an important context to determining ideal effective team structure.

What is ultimately unique about the integration of a robotic teammate on both the Mars Rover teams and my study is that this teammate ends up *doing* most of the tasks that prior studies have assigned to humans. Without robots, human only teams have had to coordinate not just their knowledge but also their skill; but with robots replacing humans in the field, what we see is that collaborative teams focus intensely on decision-making processes that determine what their robotic teammate will accomplish. Decision-making is obviously related to the communication patterns described by Urban et al. and MacMilan et al., but other studies have been able to parse out additional elements of this process as well, separate from the management of resources and the division of ability, and instead focus on managing differing sets of knowledge and values.

It has not only been observed that teams will often come to satisfactory rather than optimal decisions (March and Simon 1958), but also the multitude of factors, and multi-layered process of decision making that may lead to such a result. In hierarchical teams, team leaders rely on sub-teams and team members to make their own educated opinions about decision-making cues. As these cues are processed up the chain, it inevitably is the responsibility of the leader to synthesize this pre-processed information (hopefully reducing the workload he would have done on his own), and making the ultimate decision. The effectiveness of this decision is ultimately reliant not only on the informality of the group (how informed the group is), but also the effectiveness of the leaders' weighting scheme in determining the validity of his sub-teams and their members (Humphrey et al. 2002). Throughout this multi-level process, informational redundancy, informational saturation, and poor weighting schemes, are just a few of the obstacles to effective decision making on hierarchical teams, and furthermore a poor leader cannot make up for good staff and vice versa (Humphrey et al. 2002). In fact, team leaders are inevitably responsible for much more than just effective decision-making, and effective leaders challenge and engage their team, and are at the heart of team culture and social climate (Morgeson, DeRue, and Karam 2010).

This provides some insight into how decision-making processes would operate on collaborative teams like the Mars Rover and the hierarchical condition used in this study (described in detail in Chapter 3). However, Humphrey et al. also describe how repeated decision making sessions with the same staff led to more optimal decision over time and the Mars Rover is certainly an example of repeated sites of decision making that this study hopes to replicate. Additionally, there is the concession that "there has been far less research conducted on hierarchical decision making groups, relative to consensus decision making groups" (Humphrey et al. 2002:178) and that is a major goal of this study. Although consensus groups have been studied in the context of juries quite regularly, there are few scholarly comparisons to hierarchy-based decision-making, and observing operational differences such as the synthesis problem of hierarchy-based groups and the ability to proceed without waiting for consensus at all.

Finally, its important to address that the formal elements of team and communication structure have an important relationship in establishing, and in turn being reinforced by culture, attitude, and behavior. Culture is quite visible on the Mars Rover teams, as not only do they go through a variety of ritualized meetings and patterns of conduct but the Rover itself has becomes a cultural relic that has helped unify the team and allow them to function successfully within their given organizational structure (Vertesi 2015).

Mickan and Roger (2000) explain more holistically the characteristics of effective teams in a healthcare scenario. Shared mental models are stressed again. Effective teams not only must have shared goals and aims, but be aware of each other distinct roles and abilities, as well as have faith in and be led by an appropriate leader. The idea of faith and respect are equally important in the healthcare setting where the culture is in many ways defined by expertise, and decisions often have high stakes. Team members need to trust and be aware of the roles and abilities of their co-members in order to operate effectively. Positive social relationships between team members and effective conflict management can also be facilitated by culture as much as organizational structure.

Owens (1999) demonstrates a case study of university academic libraries desiring to enhance their culture of customer service and efficiency. As such, the two libraries studied changed from strict hierarchical organization, to flatter system of teams. In these instances, dissolving redundant layers of bureaucracy did in fact facilitate productive change. Book shelving was much more efficient and staff felt empowered by more individual responsibility and the ability to act. This in turn generated a culture of proactivity and creativity that allowed the staff to engage in more diverse roles and speak openly about the management of the library in the future.

Adding complexity to this body of research is the fact that not only does context play heavily into organizational effectiveness, but also team structure sometimes has no real influence on productivity at all. In a study of small sales teams, although team structure affected a variety of team processes, it had little effect on actual revenue. In fact, many of the attributes associated with good teams, including training, support, and open communication had no effect on actual productivity, even though they were associated with higher worker satisfaction which was *perceived* to scale with productivity, even though market growth, an endogenous variable, was more responsible (Gladstein 1984).

### 2.4 Human-Robot Teams In Context

With a firm understanding of the literature around team processes, their development, and how they enforce and are in turn reinforced by team structure, it must now be explained how the context of this literature is being changed by having robots function more often as supplementary and complimentary teammates. Robots have the capacity to be built and designed to complete a variety of tasks *with* humans as much as *for* humans, and this extends beyond the Rover that has extended our reach to Mars. Robots are being used now in manufacturing, search and rescue, military, and medical contexts, and in many of these contexts they function as more than tool, but as teammate. Even where they do function closer to advanced tool, they are affecting the way teams can and will function and organize themselves.

Julie Carpenter has focused much of her research on the growing reality of human-robot teams, attempting to parse out exactly how humans and robots can and will interact while working together. Her work goes beyond exploring efficiency and productivity, but also the social and emotional effects of working on these teams. It is clear that the Mars Rover Exploration teams have developed a significant emotional attachment to *Spirit* and *Opportunity,* facilitated by this group embodiment and relic status, that has also driven team solidarity (Vertesi 2015). Again however, because of the inherently small sample size that the rovers present, it is unclear if this emotional attachment comes with any of the drawbacks that Carpenter describes as possible on teams of similar composition (Carpenter 2013). Carpenter finds that teams that work with bomb disposal robots are often reluctant to send these teammates into dangerous situations, despite being designed precisely to do so, and experience notable levels of mental distress that may interfere with optimal decision making. Given the wide range of decisions that could lead to "success" on Mars, ranking these decisions is inherently difficult and it is even more difficult to determine if more optimal decisions existed that were ignored in favor of the Rover's health. On the other side however, one must consider that for the Mars Rover, robot health is intrinsically more important than it would be for bomb disposal robots. Bomb disposal robots are purposefully put into situations that could lead to their destruction.

It is also clear that a robot's "physiology" is partly responsible for the level of attachment and pattern of interaction (Bartneck, Reichenbach, and Carpenter 2006). *Spirit* and *Opportunity* feature a variety of parts that have been abstracted into human synonymies. The Pan-Cams have become eyes, the various geological tools have become arms and hands, and the wheels have become feet (Vertesi 2012). How would this connection and embodiment (and the team success it is responsible for) be altered if the robot were designed differently, perhaps more human, perhaps less? It has been demonstrated that when working collaboratively with a robotic partner during a simple counting task, humanoid or zoomorphic robots are praised more and punished less than human, computer, or machine-like robotic partners. Even though robots are in essence computers themselves, the differing embodiment of this intelligence affected human expectation of ability (Bartneck et al. 2006).

Satisfaction and expectation of robotic partners is not only related to form but also to function. Specifically, robotic partners in manufacturing tasks were deemed more valuable the more autonomous they were. Robots that absolved human teammates of task allocation planning were seen as more valuable and increased worker satisfaction when working with a robot. Although humans still outwardly believe human partners to be of more value than robotic ones, the more robots can decrease workload for human teammates and work autonomously, the greater team fluency, and the more efficient and effective task allocation processes become, increasing satisfaction (Gombolay et al. 2014).

The capabilities of robots and humans when they work together on teams must also be taken into account when choosing team structures. Gao, Cummings, and Bertucelli (2012) have done repeat studies of team structure and humans who work with multiple robots and compared effects of team structure on efficiency and subjective workload. Not only is the Urban Search and Rescue task similar to the function of the Mars Rover (a team controls a robot or team of robots remotely to find "victims") but the measures used throughout the study to measure performance and effectiveness inspired many of the parameters for my study as well (discussed in Chapter 3). Pooled teams, those (two participants) that could control any of 24 roaming robots, as opposed to sector teams that had a distinct set of 12, reported lower levels of subjective workload, although both teams performed relatively equally on actual task performance. It is suggested that because sector based teams had little need for explicit communication, they had to work more to prevent errors. Pooled teams were able to correct each other verbally and were able to work more with the other human teammate lessening the need to take direct control of the normally autonomous robots.

The salience of these results is still unclear because as discussed previously, communication can be as much of a cost as a benefit, and because actual task performance was similar in both groups, conclusions about optimal team structure are hazy. Additionally, similar and repeat studies have demonstrated results suggesting that sector teams actually find victims at higher rates than pooled teams, and benefit more from task automation (Lewis et al. 2011).

Finally, not only do robots allow for new types of team organization as they become valued teammates, but they can also become sites of change for existing team structures as they allow new affordances. Robot assisted surgery is one such example where team composition and structure will have to be reconsidered given the new environment. Although robot assisted surgeries allow for distributed teams and the transfer of expertise over great distances, they call for increased planning, more staff, and increased coordination well before surgery between scrub nurses and surgeons. Additionally, obstacles that will need to be addressed will be language differences, patient surgeon relations, and coordinating and synthesizing the actual views of the remote surgeon and present nursing staff (Elprama et al. 2013). Furthermore there is the issue of reliance on the robotic part of this team through which the surgeon is operating. Reed and Peshkin point out that “human-machine interfaces [are becoming] more common" and that these systems need to reflect standard human-human physical interaction. They showed that cooperative robotic systems can be designed to accurately simulate human force profiles, but that the prior knowledge of working with a robotic teammate can still hinder normal efficiency (Reed and Peshkin 2008:108).

Without a doubt human-robot teams are will become the norm as they allow for an increasingly large range of tasks, and how they are integrated into teams varies by context, by type of robot, and how teams interact both with and without this new teammate.

## 3. Experimental Design

### 3.1 Aims

The initial aim of this experiment was to simulate the context of the Mars Exploration Rover where a large team collaboratively controls a remote robot through an experimental space in order to achieve of number of exploratory tasks. As previous ethnographic work (Vertesi 2015) reveals a complex interplay between team and communication structure and how decisions are made, tasks achieved, and how the robot itself is incorporated into these processes, this research varies that very communication structure to address such questions as "For teams that work remotely with robots, how does changing team structure affect aspects of teamwork, patterns of communication, and decision making processes?" Additionally, in light of more generalized research about the role of team structure on team productivity, this study set out illuminate how the presence of a real robotic teammate changes the way varying team structures have previously affected team dynamics. Addressing the flattened, consensus-based structure of the actual Mars Rover teams versus a more structured hierarchy, will these conditions bring to light noticeable differences in how the team relates to their robot in addition to each other?

### 3.2 Sample

A total of 30 undergraduate Princeton Students were recruited via email explaining that they would be working in teams to control a robot for a variety of exploratory tasks. Subjects were awarded $10 gift certificates to Amazon for their participation. Participants were then divided into 6 teams of 5. As the experiment was team based, team composition was difficult to randomize but best efforts were made to balance genders in a 3:2 ratio on any given team.

### 3.3 Research Design

Within each team, participants were randomly assigned the role of "Principal Investigator (PI)," "Camera Technician (CT)," or "Movement Technician (MT)," where there was only one PI and two of each of the other two roles. Teams were then assigned to one of two experimental conditions, "Consensus-Based Team Structure" and "Hierarchy-Based Team Structure." Participants were given customized scripts that described their team's goals, their personal goals based on their role, as well as their decision-making conditions.

*Materials and Apparatus*

An experimental space of 15x15 square feet was built in one of the two rooms used for the experiment. This space consisted of a number of obstacles, including walls, boxes, and ramps, laid atop a grid demarcating where the robot was at any given time. Below is a representation of the map from above with the circle marking the robot's starting position, facing north (figure 2).

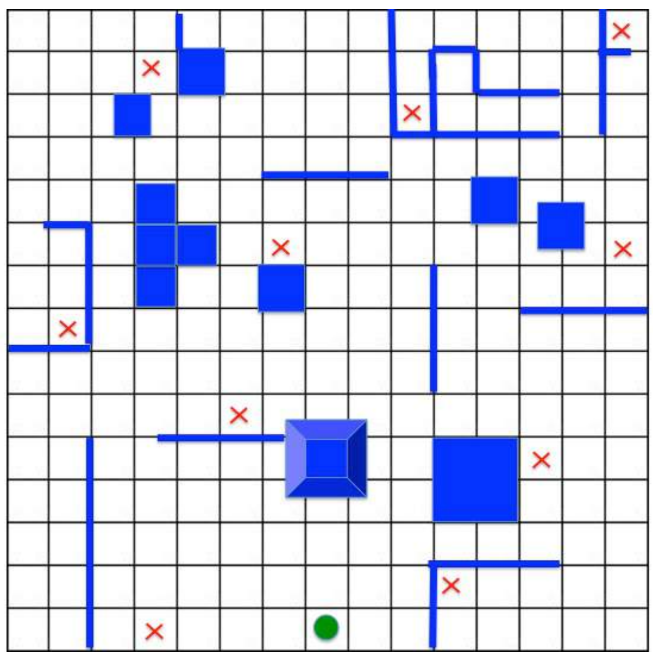

Figure 2: Map of Experimental Space

Throughout the experimental space were also 10 discoverable objects (marked by X's) that the teams would be tasked to find. The robot used was an Appbot-Link, a three-inch high, tread based "rover," with camera functionality, controlled via iphone or android app over wifi (figure 3). This robot was chosen because of its abilities to take photos, ease of use, and size.

Taking inspiration both from the conduct of Mars Rover missions and the Urban Search and Rescue Task utilized by Gao et. al (2005), the experiment consisted of keeping the robot remote from the team in control of its actions, and being fed locational information via pictures taken by the robot. Thus in a separate, adjacent room, using a computer monitor, teams would have access to a picture from the point of view of the robot's starting location (marked by the circle) facing north. Teams were also given a blank 15x15 grid on which they would keep track of their location as well as the location of obstacles and discoverable objects.

**Figure 3: "S346" - Appbot-Link**

*Experimental Procedure*

Each team was given 90 minutes to maximize exploration and exploitation operationalized as locating the position of every discoverable object as well as exploring as much physical space as possible.

At the start of the experiment, utilizing the picture from the robot's starting location, teams had 10 minutes to decide upon a "command set." Command sets were a series of commands in bulk that were to be carried out by the robot. Command options were "forward" or "backward" which moved the robot 1 foot in that direction, "turn right" or "turn left" which rotated the robot 90 degrees in that direction, "camera up" or "camera down" which tilted the camera of the robot 45 degrees in that direction, and finally "picture" which had the robot capture a photo. Command sets were given to a research assistant located in the experimental space with the robot and were executed. If at any point the command set described a command impossible for the robot (i.e. ran the robot into a wall), any following commands were not carried out and the team would alerted as to the last successful move. Any photos taken by the robot would then be sent back to the computer monitor after the completion of its entire command set. Once the pictures have successfully transferred, teams are given another 10 minutes to discuss and decide upon the next set of commands. This process continues until the 90 minutes are completed. Conversations and discussions were recorded throughout the duration of the experiment.

This "command in bulk" and delayed execution strategy was chosen because it closely resembles the actual operation of the Mars Rover. For the Mars Exploration Rover mission, the rover is too far away to control in real time. Thus at the beginning of each Martian day, or sol, Rover teams have 8.5 hours, using a previously acquired collection of information, to decide upon a set of commands in bulk they want to send to the rover to carry out. They then receive a new set of scientific information and pictures that are used in discussion to decide upon the next set of commands (Tollinger et al. 2006).

At the completion of the experimental procedure, participants were asked to complete surveys that asked them to evaluate themselves and their teammates. Participants were then debriefed and given brief tours of the experimental space in which the robot resided.

### 3.4 Experimental Conditions

Each team was assigned to one of two possible conditions that both dictated the team structure and subsequently the communication structure, as well as specified how their assigned role functioned on that team. These were "Consensus-Based Decision Making" and "Hierarchy-Based Decision Making." Scripts were carefully constructed to feel organic and explain the roles of the Principal Investigators, Camera Technicians, and Movement Technicians, in a way that fit the nature of the team structure. Under both conditions, scripts described that team members were members of an elite robotics firm (WallaceTech) testing their new exploratory robot for judgment by their superiors. The hope was that this narrative would reify the stakes of success or failure. The following are excerpts from each script explaining the experimental condition, both taken from the same section of the script (see full scripts in Appendix A & B):

*Consensus*: "As is typical of teams here at WallaceTech, they are comprised of 5 members. There is a Principal Investigator, 2 camera technicians, and 2 movement technicians. However, as we have tested that open dialogue and collaboration is the most successful for robot teams, all team members must come to a unanimous consensus for any given set of actions."

*Hierarchy*: "As is typical of teams here at WallaceTech, they are comprised of 5 members, the Principal Investigator, 2 camera technicians, and 2 movement technicians. The Principal Investigator is the ultimate decision making authority on command sets for the S346 and must balance the concerns of the sub-teams under his/her control. Camera technicians should voice their information about camera concerns and decisions to the PI. Movement technicians should likewise pass along their movement recommendations to the PI. Information and opinions can be discussed between team members but the PI will be responsible for formulating a plan to pass to the S346."

Within these experimental conditions were the three roles mentioned previously. These role assignments were chosen because they capture the role and knowledge division that is seen on many scientific teams including the Mars Rover. Typically Principal Investigators will organize and manage their sub-teams, each of which has a scientific specialty, as well as a particular area interest and set of goals they would like to see accomplished.

*Condition 1: Consensus-Based Organizational Structure*

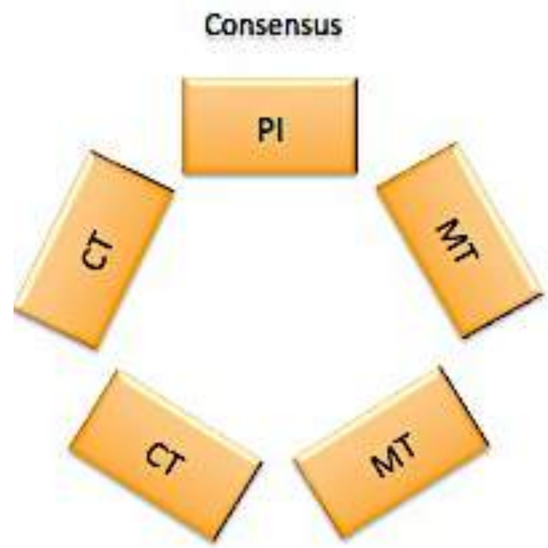


Figure 4: Seating Arrangement in Consensus Trials

In the Consensus-Based experimental condition, PI's were encouraged to facilitate discussion between members of the sub-teams of Camera and Movement technicians. As the principal investigator, they were acutely aware of the need to complete both major goals, locating every object and exploring physical space, and balance the concerns of the other team members whose goals were slightly more nuanced.

From their scripts, Camera Technicians were the most knowledgeable about the constraints and restrictions of the camera. They were told that the more pictures taken the more time it would take to transfer this information given the robot's "limited memory." They were hinted towards the efficient use of the camera as well as told to prioritize taking pictures of all of the discoverable objects.

From their scripts, Movement Technicians valued the speed and efficiency of exploration. By explaining to them to explore as much space as possible they were more incentivized to encourage more movement commands per command set, as well as more pictures for a better understanding of the environment in which they are moving.

Another critical feature of the Consensus-Based communication structure was that information was equally available to all members of the team. This was facilitated by arranging all five team members in a circle around a monitor that could be passed easily between team members (figure 4). This lateral flow of information was important because it again simulates the conditions of the Mars Exploration Rover team at NASA.

*Condition 2: Hierarchy-Based Organizational Structure*

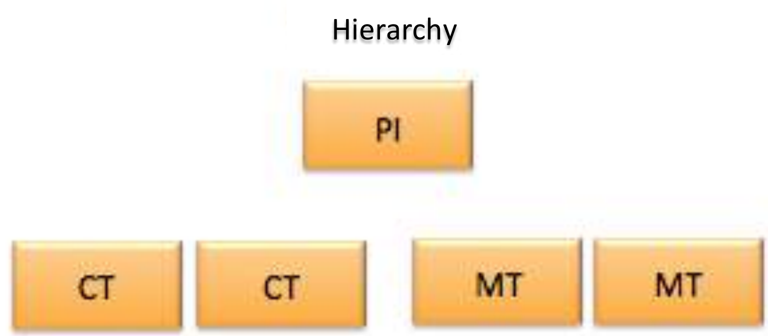


**Figure 5: Seating Arrangement in Hierarchy Trials**

In the Hierarchy experimental condition, the role of the Principal Investigator extended beyond facilitating agreement. The PI was given the ultimate decision making authority when balancing between the concerns of his/her sub-teams. He or she was to gather the information necessary for making a command set decision from both the camera and movement technicians, and synthesize this information into an appropriate decision towards the completion of the goals of locating every object and exploring physical space.

To promote the centrality of the PI role in the hierarchy, informational flow was more limited than in the consensus groups. From their scripts, Camera technicians were not only knowledgeable about the restriction of their robot teammate and valued the goal of taking pictures of discoverable objects the most, but also had initial access and control over he monitor that displayed the pictures taken by the robot. Information would have to be requested or volunteered for it to be passed along.

From their scripts, the movement technicians again prioritized speed and efficiency when exploring as much physical space as possible, but also had initial access and control of the blank team map, which they would be tasked to fill in effectively themselves. Again, this information would have to be requested or volunteered for it to be passed along.

Like in the Consensus-based group, a seating arrangement was chosen to further operationalize a more vertical flow of information towards the PI who would be synthesizing the varying sets. PI's were situated at the head of the room, while two sets of two desks faced the PI's desk (figure 5).

### 3.5 Quantitative and Qualitative Analysis

The post experimental analysis features both quantitative and qualitative measures of team performance to shed light on teamwork and team efficiency when working with a robot, as well as get an understanding for the evaluations teammates might give each other in these situations.

A series of performance metrics, many inspired by Gao et al. (2012), were tracked to appropriately measure team effectiveness in either condition. From Gao et al. (2012) I found it important to track the number of objects found within the time frame as well as the number of errors when marking these objects on the teams' respective maps. Additionally, as exploration was an equally important goal for the research teams, the percentage of total space explored was tracked as well.

In addition to these metrics, team performance between conditions was also compared by recording the number of command sets completed by each team within the 90 minute time frame, the average time it took to complete a command set for a team, the number of commands per command set, and the number of pictures taken by the team. These metrics were gathered with the hope that they would reveal differences in the decision-making processes on these teams. They give information about how long discussion lasted and how efficient that conversation might be; they provide clues as to how mature team perception and models of the physical space that their robot explored were, and even provide some clues as to how specific roles were operationalized within each team. These metrics, in combination with the survey (discussed next) and the qualitative analysis, were expected to reveal a basic profile of how teamwork and decision making processes operated in each condition on these human-robot teams.

An additional quantitative section of analysis was the post-experimental survey given to each participant. The survey allowed teammates to evaluate their own contributions as well as those of their teammates, and additionally judge their team performance as a whole and give their opinions of their robotic teammate as well. Surveys were a total of 10 questions and utilized a Likert Scale structure with answers ranging from the low of 1 to the high of 7, with the exception of questions asking about estimated average team discussions session (which could have been a maximum of 10) and ideal team size (which scaled from 1-5 with an additional "greater than 5" option). Participants were also asked about whether they believed pictures, or exploratory functions to be more paramount to team success.

All sessions were recorded and transcribed to allow a qualitative analysis to be performed on the conversations that occurred within the 90 minute trials. Basic conversational analysis units were observed such as adjacency pairs, reparative language, and conversational footing, but this was in addition to a more holistic approach that looked for unique patterns, choice of pronouns (especially "I" vs "We"), repetitions, and types of conflict and conflict resolution.

### 3.6 Hypotheses

To reiterate, this study focuses on discovering how team processes, and the development and operation of teamwork, differs when changing organizational, team, or communication structure. Additionally I have set out to answer how the differing structures tested in this experiment interact with the presence of a robotic teammate, and how the relationship to this non-human teammate changes, and how this too affects team performance. Thus my hypotheses are as follows.

*Hypothesis 1*: Consensus-based teams will, as prior research suggests, be more effective than Hierarchy-based teams in terms of exploratory performance metrics. These teams will

- Find more objects
- Make less errors in locating objects on the map
- Explore more of the experimental space

*Hypothesis 2*: Consensus-based teams will demonstrate more effective communication than Hierarchy-based teams,

- Completing more command sets
- Taking less time on average to complete command sets
- Making more commands per command set

*Hypothesis 3*: Consensus-based teams will demonstrate higher averages than Hierarchy-based teams in evaluations of their contributions and the contributions of their teammates

*Hypothesis 4*: Consensus-based teams will reference the robot more in conversation than Hierarchy-based teams, suggesting that the relationship to the robot is closer and that it is more teammate than tool

*Hypothesis 5*: Consensus-based team conversation analysis will demonstrate greater use of inclusive language (i.e. "we"), greater backup behavior, and stronger team mental models than Hierarchy-based teams

I hypothesize that the more flat distribution of knowledge as well as looser power and communication structures will allow for more effective teamwork on Consensus-based teams. As prior research shows, although very dependent on task, consensus-based teams often are more effective at establishing effective shared mental models and communicate more often, and more efficiently, needing to use communication less for the transference of information and more towards the completion of a task. Given that the task before each team is a collaborative one, I predict that a consensus-based team structures will facilitate collaboration more effectively.

I predict that the establishment of shared mental models will translate to a better-shared understanding of the exploratory space and the abilities and limits of the robotic partner. As such, I believe this will lead to greater exploration, longer command sets, and more completed command sets, given the shared access to the same body of information. Greater exploration and more command sets then leads me to hypothesize that discovering more objects will be closely linked to the amount of space explored. Additionally, I believe that Consensus-based teams, through greater amounts of discussion and a greater variety of data interpretations from teammates will lead to fewer errors when choosing a discovered object's location.

Hierarchy groups I predict to be hindered by the limited horizontal flow of information, separating those who have direct access and interpretation of the map from those with access and interpretations of the pictures. I believe that more discussion and conversation will be oriented towards interpreting data rather than future planning because teammates will not as easily be able to interpret all of the data themselves. Thus, time in each planning session will be used to bring Principal Investigators "up to speed." This will result in fewer completed command sets and likely less space explored.

I also predict that more visible power and communication structures within Hierarchy-based groups will lead to greater adoption of individual roles and goals, leading to greater conflict between team members. Principal Investigators will have greater difficulty alleviating this conflict and thus will find shorter command easier to please his or her sub-teams.

As stated in hypothesis 3, I believe that the exit surveys will reveal that consensus-based team members value their own contributions and the contributions of their teammates higher than members of the hierarchy-based teams. The consensus-based environment will promote open dialogue and will facilitate more collaboration, and in conjunction with higher predicted levels of performance success, will lead to team members finding more value in the contributions that led towards that success. Because consensus is necessary for action, team members will have to be more involved with the data and with their teammates, leading to higher evaluations.

Lastly, I hypothesize that, similar to observations of the Mars Exploration Rover teams, language on consensus-based groups will reflect the "togetherness" that will be the driving factor of their success. Consensus groups will make more prevalent use of the pronouns such as "we" while hierarchy groups I predict will still rely more on "I" and "you." Further conversational analysis I predict will reveal more examples of backup behavior and repair on consensus teams, where team members adopt the roles and make up for momentary shortcomings of their teammates. Hierarchy-based groups will not get as much opportunity to do this given their more specific roles and limited access to group information. Extending from this, dialogue reflecting and promoting team unity will extend to the robotic teammate. I predict that consensus groups will reference the robot more and that this condition is paramount to establishing a connection and an understanding of this teammate, as it seems to be for both *Spirit* and *Opportunity*.

# 4. Results

## 4.1 Performance Metrics and Team Effectiveness

| Trial | Cond. | Total Command Sets | Avg. Time Per Set | Avg. # of Command Per Set | Total Pics Taken | Avg. Number of Pics Per Set | Time Elapsed (Max 90) | % Space Explored | # Ob. Found | # Ob. Errors |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | 22 | 2:10 | 6.318 | 29 | 1.32 | 90 | 19.1% | 4 | 1 |
| 2 | C | 19 | 3:15 | 8.632 | 36 | 1.895 | 90 | 24.0% | 7 | 1 |
| 3 | C | 16 | 4:05 | 7.563 | 30 | 1.875 | 90 | 17.8% | 4 | 1 |
| | **Avg.** | **15.667** | **4:07** | **10.452** | **32.33** | | | **28.0%** | **6.333** | **1.000** |
| 1 | H | 16 | 3:31 | 11.875 | 28 | 1.75 | 90 | 36.4% | 6 | 2 |
| 2 | H | 13 | 4:58 | 11.538 | 34 | 2.615 | 90 | 24.4% | 7 | 5 |
| 3 | H | 18 | 3:54 | 7.944 | 35 | 1.944 | 90 | 23.1% | 6 | 0 |
| | **Avg.** | **15.889** | **3:54** | **9.963** | **30.11** | | | **27.4%** | **5.444** | **2.330** |

Figure 6: Performance Metrics and Averages

Upon the successful completion of 6 trials, 3 under the consensus condition and 3 under the hierarchical condition, some basic analysis was performed on the data revealing some initial suggestions about the effects of team structure (figure 6). I recognize that severe constraints on the experiment including time, availability, and access to resources do not allow the data to demonstrate statistical significance, but my hope is that the quantitative data, in addition to the qualitative data suggest certain associations that could be explored with further research.

An initial look at performance data reveals that Hierarchy-based groups on average completed fewer command sets than Consensus-based groups, averaging at 15.67 sets versus 19 completed command sets in 90 minutes. 22 completed command sets was the highest achieved by any team, and that was under the consensus condition. This appears to support the initial hypothesis that Consensus-based groups will be able to complete more command sets, due to the apparent benefits of a consensus including efficient communication and greater capacity to develop shared mental models and demonstrate conflict resolution. However, although consensus-based groups were able to complete more command sets in the allotted 90 minutes, hierarchy-based groups had command sets that were notably longer, averaging at 11.71 moves per command set. This is in comparison to consensus-based teams, which averaged 7.504 moves.

Hierarchy-based groups were able to submit and longer command sets, but they also completed these command sets in higher average amounts of time, averaging at 4:07 versus 3:10 for consensus-based teams. However it should also be stated that these times varied greatly across the trials and there were outliers in both categories. Given that, consensus-based groups attempted to keep times low and command sets short, this may suggest that achieving consensus was a more difficult process and that command sets were easier to agree upon if they were shorter. If consensus groups hypothetically worked to achieve consensus on each individual command within a set, then the shortened command sets are unsurprising. This does not support part of the hypothesis that predicts efficient communication leading to longer command sets in addition to more command sets.

Also notable was that the lengthy command sets were in fact dominated by movement based commands rather than pictures. Five of six groups took between 1-2 pictures per command on average, with only one hierarchy-based group averaging 2.6 pictures per command set. Given the photographic constraints suggested by the scripts these results are unsurprising. From here though, it is clear that hierarchy-based groups *moved* the robot much more than consensus-based groups. This is supported by the percentage of physical space covered by the robot for each group. Consensus-based groups explored an average of 20.3% of the experimental space while hierarchy groups managed to average 28.0% of the map. While not possible to call this statistically significant due to the limited trials, the difference is notable.

With a greater amount of physical space explored it is then also unsurprising that the number of objects found was higher for hierarchical teams. Objects were marked by groups on their maps when found, along with a time stamp, and given the relatively small size of the space and obvious demarcations of grid spaces, objects had to be marked in the exact box of their location to be considered correct. Hierarchical teams discovered an average of 6.33 objects while consensus teams found an average of 5 objects during the 90 minutes. . Locating more objects seems to be related with exploring more space and thus initially this information continues to oppose the hypothesis that consensus groups would be more effective given that, on average, consensus-based groups found fewer objects, along with exploring less space.

However, what appears to support my initial hypothesis is that consensus teams appeared to generally make fewer errors, all three groups only making 1 error apiece, while hierarchy groups on average made 2.33 errors, with team two making a total of 5 errors in object placement. Initial observations of qualitative data suggest that consensus-based teams were very concerned with detail and certainty throughout the mapping process, as compared to hierarchy-based groups which, as supported by the lengthier command sets, had either greater shared models of the experimental space or were more willing to make riskier decisions based on approximations. The shortened command sets, in addition to the few errors made by consensus groups, may suggest that consensus was easier to achieve in planning sessions if action was taken toward confirmation rather than exploration.

It must be said now that this initial report on the results of the experimental trials is done with my starting hypotheses in mind, assuming that the data would divide itself in the predicted bipartite fashion along conditional lines. However, when the performance metrics are viewed together, rather than revealing two distinct groups, one consensus-based and one hierarchy-based, there appear to be in fact three distinct clusters (figure 7). Not only does this appear in the quantitative data based on performance, but also observations and scripts from trials reveal that one group from each condition experienced a notable decoupling of the formal and informal processes that make up their respective conditions. Both the second consensus-based trial and the third hierarchy-based trial appear distinct from groups in their shared condition, but very similar to each other across quantitative and qualitative data. When dividing the data into three sets, the similarities between the remaining two teams in each condition also appear more pronounced. Again, this appears to be backed up by qualitative analysis as well as exit survey data. Across conditional lines the survey shows much less remarkable trends than when treated to show three distinct groups. This ultimately begs the question about what was operationalized differently within groups that were given a certain set of restraints to work with when acting as a team. Did a consensus-based group in fact organize themselves closer to a hierarchy or vice versa? And how did these groups manage to decouple formal and informal processes of their conditions. The following chapter presents observations and results based on the recordings from the 6 trials, demonstrating two groups of tightly coupled hierarchical arrangement, two groups of tightly coupled consensus-based arrangement, and two groups of loosely coupled arrangements, one in each condition (figure 8)

| Trial | Cond. | Total Command Sets | Avg. Time Per Set | Avg. # of Command Per Set | Total Pics Taken | Avg. Number of Pics Per Set | Time Elapsed (Max 90) | % Space Explored | # Ob. Found | # Ob. Errors |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | 22 | 2:10 | 6.318 | 29 | 1.32 | 90 | 19.1% | 4 | 1 |
| 3 | C | 16 | 4:05 | 7.563 | 30 | 1.875 | 90 | 17.8% | 4 | 1 |
| 2 | C | 19 | 3:15 | 8.632 | 36 | 1.895 | 90 | 24.0% | 7 | 1 |
| 3 | H | 18 | 3:54 | 7.944 | 35 | 1.944 | 90 | 23.1% | 6 | 0 |
| 1 | H | 16 | 3:31 | 11.875 | 28 | 1.75 | 90 | 36.4% | 6 | 2 |
| 2 | H | 13 | 4:58 | 11.538 | 34 | 2.615 | 90 | 24.4% | 7 | 5 |

Figure 7: Performance Data Rearranged to Show Clustering

## 4.2 Tightly and Loosely Coupled Team Structures

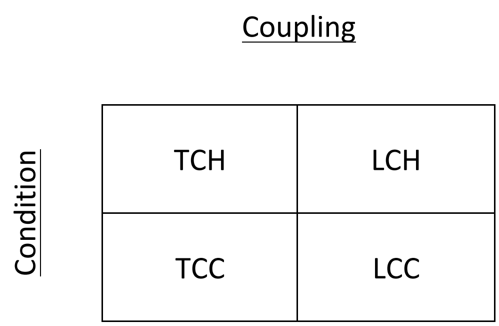


Figure 8: Tightly and Loosely Coupled Conditions

Meyer and Rowan discuss organizational "gaps in formal structure and actual work activities" (1977:341) and describe these organizations as loosely coupled, in order to maintain efficiency in the face of uncertainty, or to address a lack of efficiency embedded in the formal structure. As such, they allow for the maintenance of formal structure while their actions do not necessarily reflect it (Meyer and Rowan 1977). An example once given to me would be the knowledge that one must in fact go to the secretary in a highly bureaucratized organization to achieve anything rather than going to one's superiors. Here the structure of bureaucratic superiors, executives, VPs, etc, is maintained, while actual work processes happen at the fringes of this structure. The opposite of course would be tightly coupled organizations, which adhere strictly to the processes of their designated structure. In the previous example this would be akin to sending information all the way up and down the bureaucratic chain as intended, rather than other forms of subversion or "alternative" means of work.

*Tightly Coupled Hierarchical Teams*

| Trial | Cond. | Total Command Sets | Avg. Time Per Set | Avg. # of Command Per Set | Total Pics Taken | Avg. Number of Pics Per Set | Time Elapsed (Max 90) | % Space Explored | # Ob. Found | # Ob. Errors |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | H | 16 | 3:31 | 11.875 | 28 | 1.75 | 90 | 36.4% | 6 | 2 |
| 2 | H | 13 | 4:58 | 11.538 | 34 | 2.615 | 90 | 24.4% | 7 | 5 |

Hierarchy teams 1 and 2 comprise the first cluster that I am labeling as tightly coupled. Across the performance data they appear very similar having a low total number of command sets, 16 and 13 respectively as well as near identical average numbers of commands per command set at 11.875 and 11.538. These two groups also explored the most space, with hierarchy team 1 exploring 36.4% of the map, and made the most errors in object placement.

Within each of these trials I observed that teams adhered more strictly to the given hierarchy than did the third team, which I am labeling loosely coupled, and will discuss along with the loosely coupled consensus team. These tightly coupled hierarchies frequently invoked their title and roles and often made decisions based on how knowledge was divided among these roles. For example, one exchange during the first hierarchy trial had a Movement Technician state what a command set would be and that it ended with a picture but then followed with "Camera technicians, you decide how you want that photo taken."

Along with the invocation and adoption of the roles described to participants in their scripts, throughout the 90-minute time period a level of expertise appeared to have developed on sub-teams, and role functions became more specific. An additional result of this was that the way conversation and decision-making took place reflected a degree of trust, or at least deference, between team members. PIs in both scenarios took recording and making clear finalized command sets to be their primary function, movement technicians became adept at mapping and explaining their maps, and camera technicians were the most trusted to make assumptions and observations based on visual data. On Hierarchy team 1, one memorable exchange between the Movement technicians and Camera technicians went as follows:

> *MT-1: Can we identify where they are from right here [CT name]?*
> *CT-1: Hh hang on, yeah I think we can. So roughly 4 right?*
> *MT-1: You can do diagonals. Just do diagonals. How many diagonals from the square in front of us?*
> *CT-1: 1,2,3,4*
> *CT-2: 5…and one forward after that? Or is that 2 forward?*
> *MT-1: 5 diagonals and one forward.*

In this instance the movement technician defers to the camera technician about the distance of an object. The movement technician who experimentally is given a better understanding of the geometry of the map reminds states that diagonals would be most effective, and then the camera technician answers in those implicitly agreed upon terms. The movement technician does not request to see for himself or make his own judgment (as will be observed often on consensus-based teams) and summarizes the two judgments of the camera technicians into one confirmatory statement.

Not only were members of sub-teams most likely to give the most definitive information or opinion on their respective information sets, but also sub-teams in many instances throughout both of these trials acted independently, often to clarify or go over data with their immediate teammate. In hierarchical team 2 for example, during one period between command sessions planning, a movement technician asked his teammate about a blank space left on a map while concurrently the camera technicians discussed the nature of an apparent obstacle, the "ramp" in this case.

Both of these teams acted quickly to geo-locate themselves within the space and identify nearby obstacles and objects in order to move onto movement planning. These plans often had the following plans in mind as well, suggesting that tightly coupled hierarchical groups were more oriented towards long term planning and goal completion. Hierarchical team 2 adopted the language of cardinal directions to make it easier to communicate the robot's orientation and where they wanted to go. The movement technician for this team would also start each command set planning sessions with “ok so we are here [indicate to map] facing [north/south/east/west]” hopefully orienting his team immediately so that discussion about action could begin.

As part of this combination of deference to, or trust of, those with more expected knowledge, an emphasis on long term planning, and speedier establishment of geographic elements, tightly coupled hierarchical teams appeared more comfortable working with approximations and educated estimates. Phrases such as “at least” suggest that teams intended to maximize the limits of the robot’s movement as much as possible. These teams also were more willing to ignore obstacles and information on which a level of certainty could not be reached as long as it didn’t disrupt the current or alternate movement plan. For example, discussion about an obstacle “3ish” spaces to the right would be tabled if it would not disrupt current plans to move straight, the assumption being it would reveal itself later. Moreover, even when there existed disagreement between team members concerning some of these uncertainties, the basis of the hierarchical condition allowed them to take action without addressing this. At one point for example, a camera technician on team 2 stated that he remained unconvinced of an object’s placement and the robot’s location, but conceded that the rest of his team was comfortable with their next set of commands regardless. All of the above help explain the long command sets focused on physical exploration attributed to hierarchical teams in the previous section.

Additionally, these teams featured high levels of discussion and communication throughout all of the 90 minutes, talking both during the allotted 10 minutes for planning a command set, and the time that existed in between, while the robot executed the given command set. These conversations, as opposed to consensus-based teams, featured remarkably low levels of interruption in comparison, and repetition and reiteration were much less frequent. As will be discussed for consensus-based teams, repetition was often associated with uncertainty and for clarifying missed, or misunderstood, information. Often this would result in "over-clarification" and "over-confirmation" with multiple team members finishing off the question-answer response pair. On hierarchical teams, repetition or reiteration was usually only attributed to Principal Investigators repeating the commands that were stepwise dictated to him/her as confirmation that they have been added to the set, and even this characterized only a portion of sub-team-team leader interactions. It should be noted that this was more noticeable on the second hierarchical team whose PI was notably unenthused with his role and asked me multiple time to change his randomly assigned role.

Lastly, other shared characteristics of these tightly coupled hierarchical groups are their pronoun usage and recognition of success. Both of these teams used "we" more than I had originally predicted and not dissimilarly from some of the consensus groups discussed in the following section. Although "I" was prominent when expressing ideas or opinions, such as "I think this object is here," "we" was used when implicitly referencing the robot's capabilities. "We are facing this way," "We took 3 pictures," or "I think we should go there" are common phrases used on each team. "I" was featured more on hierarchical teams than in consensus teams but the use of "we" as well suggest that team as a whole owned their success and ultimately acted together, with their robotic partner. This is incredibly akin to Vertesi's observation that the "rover's body is a body politick too" (Vertesi 2015:179) and that the use of "we" subtly aligns the team "alongside the actions of the robot" (Vertesi 2015:180). Concerning the recognition of success, both of these teams also made explicit reference to the fact that they felt they were completing the task successfully, even though they had no real indication of whether this was he truth. One exchange on hierarchy team 2 reads:

> *MT-1: So far we have 5 objects of interest*
> *CT-2: That's not bad*
> *MT-1: Which is not bad*
> *CT-1: Unless there are 30 f them [laughs]*
> *MT-1: But I'm proud that we found the object; that was a big gamble*
> *CT-1: I like how at the beginning we were trying to figure out every single square and now we're just like fuck it. It looks clear.*

Here this team is aware of the risks their strategy involves and the successful payoff. They also recognize that finding 5 objects logically seems like they are being effective and completing their task successfully. Consensus teams often knew that their command sets would complete successfully given their short length, and were often frustrated at the lack of explicit indication of positive progress.

These teams were tightly coupled because the hierarchical condition was embedded in most of their processes. Sub-teams were able to act independently from each other with the data they had access to and subsequently became the experts concerning this data. From here, information was transferred between teams without further discussion or analysis in order to synthesize it into a command set that would be passed to, and confirmed by, the principal investigator. These teams didn't operate in a way that sought consensus, avoiding when possible the stepwise discussion of steps and ignoring rather than discussing team uncertainties in order to progress.

*Tightly Coupled Consensus Teams*

| Trial | Cond. | Total Command Sets | Avg. Time Per Set | Avg. # of Command Per Set | Total Pics Taken | Avg. Number of Pics Per Set | Time Elapsed (Max 90) | % Space Explored | # Ob. Found | # Ob. Errors |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | 22 | 2:10 | 6.318 | 29 | 1.32 | 90 | 19.1% | 4 | 1 |
| 3 | C | 16 | 4:05 | 7.563 | 30 | 1.875 | 90 | 17.8% | 4 | 1 |

The next cluster that appeared from the whole of the performance data is what I am labeling tightly coupled consensus-based teams. Like tightly coupled hierarchical teams, these team operated within, and fully embraced the restrictions (or lack thereof) of their given condition. These teams did not subvert their condition in lieu of a discovered more efficient way of operating and thus again, the formal and informal processes associated with a consensus-based organization were observed in conjunction.

These two trials, consensus teams 1 and 3, although different in the amount of completed command sets had very similar average number of commands per command set, 6.3 and 7.5, and amount of space explored with those command sets. In fact at 19.1% and 17.8% respectively, these two teams explored the least amount of space, which is the first striking similarity given their trial condition. Second is that both of these teams discovered 4 objects within the experimental space, making them both the least successful in terms of this metric. However, as was seen by all 3 consensus-based teams, they only made 1 error in object placement on the map. Of course my observations of their discussions were also very similar and it begs the question as to what about the consensus condition, when visibly adhered to, led to these results.

What is most obvious from the transcripts of both of the trials was the marked amount of interruptions, reiterations, and multiple confirmations that occurred during the discussions. It iss not unfair to say that these two groups conducted their team in a more "free-for-all" style. Take this one exchange from consensus team 1 for example:

> *MT-1: Ok we're facing this way right?*
> *CT-1: We're looking at this thing. So we need to turn right cuz we are looking at this, we are on this bloc—*
> *PI: Where is the object thing—*

*CT-1: Turn right…*
*MT-1: This is the first picture—*
*CT-1: yes—*
*MT-1: We took the first pictu—*
*CT-1: And then we went forward 2 and turned right*

Not only is nearly every sentence cut off, but also many of these interruptions are not predictive. It's evident that these interjections are often unrelated to the statements of the previous speaker and that every team member is trying to establish a basis of knowledge for him or her-self. Its unclear as to whether the group is most concerned with location, movement, or even recognizing the order in which pictures were taken. This is perhaps unsurprising given that in a consensus-based group, members are responsible for all types of knowledge and any question directed at the group should technically be answerable by anyone else in the group given the shared information. However, many times this led to asynchronous discussion and immediate versus future goal planning.

The other features mentioned were the multiple repetitions and confirmations demonstrated by both groups. In consensus team 3, the following exchange was concerned about the location of an object:

*MT-1: So like 1, 2, 3, 4, 5*
*MT-2: 3... uh yeah*
*PI: It looks like 4 to me oh yeah 5 yeah, and then what's that yellow thing?*
*MT-2: Yeah some another like...*
*PI: Is it an object?*
*MT-2: Probably*
*MT-1: Yeah it's an object. I feel like there's a line there*
*MT-2: Mhm*

*PI: Yeah*
*CT-1: 1, 2…3?*
*MT-1: Alright lets just do that, so, we're here.*
*CT-1: Yeah*
*MT-1: 1, 2, 3--*
*CT-1: Wait other way though, this is looking towards…oh nevermind*
*MT-1: Yeah 1, 2, 3*
*MT-2: Yeah*

In this exchange we see multiple team members counting alloud the same 3 spaces and multiple team members uttering confirmatory phrases like "yeah," "probably" and "mhm." Additionally the response pair of "Is it an object?" and "Yeah it's an object" is interesting in the way it repeats the question as an answer instead of just a remark in the affirmative. It appears to demonstrate and indication of "extra" confirmation, and a way of indicating that at least those two team members on are the "same page." This exchange actually continues for longer with these patterns repeated. Soon the group begins to believe the object is *four* spaces away and the counting aloud from every teammate continues.

From group 1 we have another example of a brief sample of the discussion that demonstrates similar qualities:

*CT-1: So forward 1 left 1*
*MT-1: Wait*
*CT-2: Forward 4?*
*MT-1: We're facing that way*
*MT-2: We're looking that way*
*PI: We're facing forward? Yeah we're still facing forward*
*MT-1: Ok so forward 1*
*CT-1: Forward 1, left 1, forward 4*
*CT-2: yeah*

Here again we see the triple repletion of teammates confirming where the robot is currently facing, and while unclear as to whether they are stating this for themselves or for the team, it suggests to me that achieving consensus goes beyond agreeing on a submitted command set. The group implicitly requires that its members agree on all aspects of their task including orientation, location, and object placement. It is still unclear however as to whether this is an indication of well established shared mental models, or the lack of those mental models, requiring them to state aloud multiple times how they align with the group.

Perhaps another indication of the limited group understanding of the task at hand, is that not only are repetitions present adjacent to each other in the discussion transcripts, but there are multiple examples in either group of a team member asking a question (again often concerning robot orientation), having it answered by a member of the team, and then later within the same 10 minute frame having that question asked again by a different member and repeating the process.

These reiterations were also present during the discussion over actual commands, however there is a slight difference between consensus groups 1 and 3 in this regard. Although accurate to say commands were repeated multiple times in both groups, on team 1 these commands came from many team members simultaneously, with teammates either finishing (or interrupting) each others' sentences or adding the previously stated command to their won sentences. Group 3 rather had teammates express commands they were interested in, or general end locations. Teammates would then as before indicate multiple times they were in agreement with that piece of a possible command set. Another teammate would then read aloud the now completed command set to the group for approval, and then read it again to the Principal Investigator who still adopted the task of writing the command sets down (although note there was no indication that he had to take on this job).

The processes mentioned above help to support why shorter command sets and fewer discovered objects were present in both of these tightly coupled consensus-based teams. It is clear that greater effort and multiple forms of agreement were necessary for teams to confirm that they had reached consensus. In fact, because this consensus was so important for a team's success, it may be possible that these groups discovered objects but were reluctant to mark them down at all if it would amount to uncertainty. My initial hypothesis about consensus-based groups would be that they featured greater, and more efficient communication, and ultimately this is a difficult metric to parse out. Many of the completed command sets for these teams were completed very quickly, but alternatively they featured few command sets. Repetitions would also point towards inefficiency.

Conversely a point of possible support would be the presence of backup behavior or teammates correcting and supplementing the jobs and roles of other teammates. Because of the open forum structure of these consensus groups, there were more opportunities for team members to correct each other and interpret data in different ways from each other. Perhaps over the long term this would result in greater amounts of trust and more established mental models, but in the short term of this experiment, it is unclear as to whether this ability positively, negatively, or neutrally affected team effectiveness.

Lastly, observations of these tightly coupled consensus-based teams do appear to positively support one aspect of my hypotheses and that is team language, with special emphasis on pronoun usage again. While in the tightly coupled hierarchical groups I observed a balanced mix of "I" versus "we" throughout the discussion initially not supporting my hypothesis of an "I" based dialogue; these consensus teams were strongly dominated by the use of "we" throughout the experimental trials. The following sample from consensus team 1 is an example of this:

> *CT-2: **We** wanna turn right and take a picture, cuz that's the edge*
> *MT-1: Yes but where, how many forward should **we** go?*
> *PI: **We** need to get to that other thing **we** saw*
> *CT-1: **We** already have a picture of that don't **we**?*
> *CT-2: Go back to the picture where **we** have the other thing*
> *MT-1: No what im saying is **we** want—*
> *PI: **We** need to find that other yellow thing*
> *CT-1: So look where the other yellow thing is. Right there. Ok so **we** don't need a photo to the right. **We** need to get there*
> *MT-1: Wait that's right next to the big box, which is that beast right there.*

As one can see from this sample, the use of we is used for nearly everything, describing both present achievements and future planning. By using such phrases as "we need," every individual member of the team is essentially a proxy for the whole team underscoring the fact that consensus is the necessary condition for action and success. However, interestingly enough, these two teams, especially consensus team 1, generally spoke negatively of, (or even questioned the real life usage of) their robotic teammate, making many references to its irrelevancy and limitations (note however, that although more outspoken than teams in other conditions, opinions concerning the technical limitations of the robot do not notably differ from those teams based on survey data as discussed in the following chapter). In the spirit of the game however, these teams continued onwards. I posit that this language is a result of the robot not possessing a specific role on the team the way it might implicitly have in a hierarchical condition. One can speculate that because the robot cannot actively contribute to the consensus, it is reduced more to a tool rather than a teammate actively engaged in the completion of the task. It is unclear, yet interesting to think about, whether this affected the productivity and effectiveness of these tightly coupled consensus teams.

*Loosely Coupled Teams*

| Trial | Cond. | Total Command Sets | Avg. Time Per Set | Avg. # of Command Per Set | Total Pics Taken | Avg. Number of Pics Per Set | Time Elapsed (Max 90) | % Space Explored | # Ob. Found | # Ob. Errors |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | C | 19 | 3:15 | 8.632 | 36 | 1.895 | 90 | 24.0% | 7 | 1 |
| 3 | H | 18 | 3:54 | 7.944 | 35 | 1.944 | 90 | 23.1% | 6 | 0 |

The last cluster that emerged was the grouping of hierarchical team 3 and consensus team 2. My real time observations reflect that I initially found each of these teams operating in ways that reflected the opposite condition. The consensus team established a hidden power dynamic and “experts” emerged from the group, while in the hierarchical team the group experienced a highly lateral flow of information, acute attention to detail, and shortened command sets and a desire for more immediate gratification. While I was initially concerned about how to discuss this phenomenon I was surprised to find that the quantitative performance data seems to support that there is a relation between these two groups. Both of these groups completed a similar number of command sets, 19 for consensus team 2 and 18 for hierarchy team 3 and the average number of commands were close as well 8.632 and 7.994 respectively. These two numbers also fall between the two other clusters in which hierarchy groups featured a large average number of commands per command set and consensus groups featured notably fewer.

Furthermore, both of these teams achieved exploratory success reminiscent of the tightly coupled hierarchical teams, both exploring greater than 23% of the map and finding 7 and 6 objects respectively, but also demonstrated a low number of object placement errors, 1 and 0 respectively. Thus their “middle of the road” approaches, in respect to condition, seemingly allowed them to reap the benefits attributed to either organizational structure. However, both of these teams successfully formally appeared to function within their given condition. The consensus team still needed a unanimous decision to move forward and the hierarchy team was limited in who could manipulate specific data. Thus, the ability of these teams to supplement their formal organization with informal processes reflecting the opposite condition has led to the labels loosely coupled hierarchy and loosely coupled consensus.

The second consensus group was first notable for the few interruptions that took place throughout the discussions. Not only were decisions periods organized but also most notably, hidden expertise emerged from the apparent consensus. One movement technician on the team came forward as the one to record command sets and wrap up discussion. The consensus structure at its core allowed him to adopt this role for himself, but from this point he became the go-to. Most notably from here was that command sets were rarely repeated or reiterated to the group. The group would discuss and this movement technician would record in real time, only repeating if he was asked to do so. The following sample is a good example:

> *MT-1: Well if it is then we cant get in there*
> *MT-2: Well if its blocking only this square there could be an object here that we missed CT-1: But we wouldn't be able to get to it anyway*
> *MT-1: No we're here*
> *CT-2: We're gonna go here, and look down*
> *MT-2: Ok. Alright*
> *CT-2: Alright so we wanna go...*
> *MT-1: We need to turn right twice*
> *MT-2: We can just go backwards 2 and then turn right*
> *MT-1: Can we go back?*

*CT-1: Is there a backwards command?*
*MT-2: Reverse yes*
*CT-2: Ok*
*MT-2: So backwards 2, right*
*CT-2: Right photo. And then do we wanna go left, left photo too?*
*CT-1: Yeah*
*MT-2: Sure*
*MT-1: Yep*

This sample actually reveals a lot. This group clearly features some of the multiple confirmations like other consensus groups but little of the repetition of commands. In fact they are only repeated once and team members on this team are able to add to, and edit the unfinished command set without reiterating the whole set again. From here, MT-1 writes what the group has clearly already agreed to without seeking further consensus again. Also note in this sample the dominance of the pronoun "we," but also the emergent expertise associated with specific roles. Movement technicians are evidently more aware of the robot's abilities, as well as have a better understanding of the map and it current orientation. This however, while reflective of a subtle hierarchy, does not result in a reduction of the use of "we," indicating that the collaborative nature of the consensus is still preserved and is evidence for loose coupling.

An additional tool that this consensus-based team utilized to avoid the restrictions of consensus was a variety of phrases that came off as neutral or ambivalent. Many scholars have discussed the phenomenon of "hedging" within conversational analysis where speakers purposefully use vague language to “personalize or otherwise mitigate the force of what they say,” often because they wish to avoid sounding “definite or authoritative” (Iwata 2005). Using phrases such as “we might as well,” “whatever,” or even a dejected “sure” allow the team to achieve apparent consensus even if team members do not truly agree or have no opinion at this time. The group is able to proceed instead of having to work through possible conflict or fake enthusiasm for a given command set. The following sample is a good example of this process:

> *MT-2: So what are we doing here? Its gotta be unanimous*
> *MT-1: So we’re going this way. So we’re following our sight line basically so we don’t hit any obstacles*
> *MT-2: And ending in the corner?*
> *MT-1: And ending in the corner*
> *MT-2: ok*
> *MT-1: Picture up and picture…*
> *MT-2: It’s fine. It’s wasteful but its fine*
> *CT-1: Whatever*
> *MT-2: Let’s do it*

Here we get to see the soft separation occurring between sub-teams. The movement technicians in this group are essentially planning this command set given their more complete knowledge of the map. The camera technician, perhaps unclear about the command set decides to just defer to their opinion and demonstrates her consent with “whatever.” Even the other movement technician voices some displeasure with the command set, although it is unclear if it is the movement or the number of pictures, but decides it is not worth halting further progress. He says, “it’s fine” and “let’s do it” to avoid any possible hold ups due to conflict.

Another similarity between this consensus team 2 and the tightly coupled hierarchies is the ability to operate under uncertainty. While the tightly coupled consensus teams may have been slowed by their attention to detail, this group was comfortable not knowing and proceeding.

> *CT-2: 1, 2, and a half. You think it stars on the half?*
> *MT-1: Yeah*
> *MT-2: There's at least something we don't know. It looks like it at least extends to the left a little bit where we can't tell how far it is*
> *CT-2: So do you wanna say 2 blocks for now?*
> *MT-1: Yeah*

The group recognizes that with the information it currently has it won't be able to with certainty place an obstacle on a map, but they settle for approximations. At points, this team will choose to ignore some information altogether as well. Later in this discussion session a team member will ask about an object or obstacle and another responds "It's one of the shit over here we don't care about." Ultimately by limiting the amount of times a consensus needs to be taken, and settling for approximations, this group is able to complete longer command sets (compared to tightly coupled consensus groups) and explore greater areas of the map. This is also further evidence for the gap that exists in this group between the formal and informal processes of a consensus-based team.

The third hierarchy team was loosely coupled from its formal structure oppositely from the previous group. Where the loosely coupled consensus group managed to establish hidden lines of separation between the roles and reduced the number of times consensus was necessary, this group subtly broke down the lines separating the roles within the hierarchy, and its attention to detail and more collaborative nature called for more situations that desired a consensus.

Movement technicians were of particular interest. Movement technicians on this team had a very detailed process for creating the map using a variety of line types and a mixture of pen and pencil to demarcate objects and obstacles. They, because of the formal structure of the team, were also the only ones permitted to edit the map. However, throughout the experiment, the map was passed frequently to the camera technicians and the PI who collaboratively commented and gave instructions to the movement technicians about what to add to, or how to edit the map. Two patterns emerged from this behavior. First, the map itself became a site for discussions and disagreement that was often resolved through a consensus given the movement technicians' close attention to detail and specific coding scheme. Second, the movement technicians' close attention to detail was already reminiscent of tightly coupled consensus groups. Because of this, this group became less inclined to work with approximations and this likely led to shorter command sets, compared to the tightly coupled hierarchical group.

The hierarchy did show through at times throughout the trial however. Often with the arrival of new pictures, movement technicians could initially work separately on their map before giving it over to the group effort while other team members discussed future planning. Additionally, movement portions of command sets were not discussed for very long in any given planning session. In this case, team members were often willing to defer to whoever emerged as expert at that given time, although whom that was would vary. The way this affected command sets though was unexpected. This was the only group not to turn left and right and take pictures in both directions as their first completed command set. Instead this group took a straight shot forward to end up atop the pyramid positioned ahead of the robot. This is also surprising in that groups, especially tightly coupled consensus ones, often take some time to discuss the nature of the pyramid and question whether it is climbable but not here. So again, this is evidence for a "mixed methods" type of group that I am calling loosely coupled.

The apparent success of the tightly coupled hierarchical groups may suggest that it is the aspects of a hierarchy that are allowing the loosely coupled groups to find comparative success, but more data would have to be collected to expand on these hypotheses. However, beyond the performance parameters that originally illuminated the tight and loose coupling distinction, the post experimental survey data, asking questions concerning thought and opinions on qualitative valuation of member contribution, team trust, and robot effectiveness, continues to reveal a distinction between tightly coupled consensus groups, tightly coupled hierarchy groups, and the two loosely coupled groups together.

## 4.3 Team Trust and Value Within Tightly and Loosely Coupled Teams

The survey data revealed surprisingly little when calculating basic averages across answers between the initial two conditions. Questions asked participants about how they valued their own contributions, how they believed their contributions were valued by others, about their trust in their teammates, and trust in the robot. Questions utilized a Lickert scale typically ranging from 1 to 7. In most if not all of the 10 questions none of the differences attracted any attention. However, while analyzing the data I decided to run it through a series of treatments to see any new patterns appeared.

The first treatment (Appendix E) was to separate out Principal Investigators from members of sub-teams, while maintaining the initial divide between experimental conditions. This revealed a few results, which when considered, were not unexpected. Although most questions again had few differences, members of sub-teams in the hierarchical condition had notably more trust in their principal investigators, averaging at a 5.58 versus a 4.67. Given that principal investigators are typically the authoritative figure on these teams and essentially act as the direct communication to the robot, it makes sense that camera and movement technicians would implicitly trust their PIs more.

Additionally, sub-team members in hierarchical conditions found their robotic teammate less limited than in consensus-based groups. This may be because camera and movement technicians in hierarchy-based groups are inherently more concerned with only a specific function of the robot. The limitations of the robot appear to be fewer if these team members are restricted in the amount of functions they need to be concerned with.

The second treatment of the survey data was aligned with the previous treatment. Sub-team members were again separated from Principal Investigators across conditions, but now also separated for individual role (Appendix F). Camera technicians trusted the PI more in the hierarchical condition 5.67 versus 5.17. This was consistent for movement technicians as well with an even more notable 5.50 versus 4.17. Additionally, Camera Technicians in the hierarchical condition believed that their teammates valued their contributions more than in the consensus condition at an average of 5.50 versus 4.67. This again may be explained the same as before, in that in the hierarchical condition, the separation of roles forces teammates to trust each other more as they emerge as expert at their function. This of course was similarly supported by the conversational data in the previous section, where I argued that the lack of interruptions, and the ability to allow sub-teams to operate independently reflected this heightened trust. Camera technicians may also believe their teammates valued their contributions more on hierarchical teams because of the essential nature of their function, interpreting photos, and being the initial people with access to these interpretations because of the formal separations.

However, what is most surprising about the data does not appear when dividing the survey results along conditional lines, but instead when using the same tri-partite divisions that the performance data revealed. After observing the potential for three, rather than two clusters in the performance data, the survey data emerges as three clusters when averages are taken along these same divisions (Appendix G).

Five questions from the survey showed a difference between tightly coupled consensus-based teams, tightly coupled hierarchy-based teams, and the group of two loosely coupled teams in either condition. These questions were how individuals felt their teammates valued their contributions, trust in the principal investigator, trust in ones other teammates, ideal team size (from 1 to greater than 5), and how well an individual believed his/her team adjusted to the limitations of the robot. In each of these questions however, what emerged were similar averages for both of the tightly coupled conditions, and a notably different average for the loosely coupled cluster.

It should be noted though, that many of the averages for the tightly coupled hierarchy groups are surprisingly low. The principal investigator of hierarchy group 2 was not very enthusiastic about his/her participation in the experiment and, although this person performed the role successfully, was unconvinced of its necessity. Thus, this individual's answers on the survey are uncharacteristically low. This was revealed both through the principal investigator asking me if I could reassign his role at the beginning of the trial, and a brief post experimental interview, in which he revealed his skepticism over the importance of the principal investigator, and expressed doubts over his own leadership ability. Therefore, my observations are more focused on what was different between tightly and loosely coupled groups that caused loosely coupled groups to stand significantly apart from tightly coupled ones, rather than explaining a difference between both types of tightly coupled groups.

For how an individual believed his/her teammates valued his/her contribution, tightly coupled consensus and hierarchy teams averaged at 4.6 and 4.9 respectively while the loosely coupled group averaged only at 3.9. For trust in the principal investigator, 5.8 and 5.1 versus 4.8 (and note that 5.1 is affected by the outlier answers of the unenthusiastic PI). For how much an individual trusted the rest of their teammates, the averages were 6.1 and 5.7 as compared to 4.9. Trust and teammate value are lower across the board in loosely coupled groups. This may be because in both of these teams, the social order is inherently unclear. Because they are working outside the formal boundaries of their given condition, it becomes harder to pinpoint the exact function of a given teammate, how one is allowed to interact with that teammate, and how one operationalizes one's own role on a team as well.

This rationalization may also explain the lower averages for ideal team size and team adjustment to robotic limitations. While tightly coupled consensus and hierarchy-based teams answered 3.5 and 3.3 people per team respectively, loosely coupled ones averaged at 2.6. Without the ability to effectively see social order, fewer teammates would inherently reduce this confusion. Team adjustment was at 5.8 for both tightly coupled groups and only 5.3 for loosely coupled ones. Similarly, where social order is unclear it would appear harder to adjust to any robotic limitations. Where normally robotic limitations would either be negotiated equally among team members or the primary concern of a sub-team, the loosely coupled teams fall in between these two methods.

### 4.4 Team and Robot Relationship

I'd like to turn now to my observations of how the teams in this study established a working relationship with the robot. Although there is nuance to each team, it undeniable to say that any team was unaware or established no relationship with the working partner that was the "S346." Additionally, within the three clusters of teams, there exist some similarities in how this relationship was expressed.

As discussed, in every group the use of "we" was present to some degree. "Which way are we facing?" and "We took a picture here" are just some examples of commonly used phrases. Rarely was the robot ever referred to as "it" "he" or "she." For me this could suggest a few different results. On the one hand, the teams could be including the robot in the success and action of the team. The team worked together to make the S346 take a picture; it did not do it on its own. It contributed to this effort as a team member. On the other hand however, as the rover was rarely recognized for its individuality, it could be that the teams give no credit to the ability of the robot and it is more tool than teammate. There could also exist in between states to either of these two sides.

Moreover, at some point during each trial each group brought into question the robot's existence. Although each group was given a demonstration of the robot and its capabilities before the trial, due to the fact that the team had no other contact with the robot, some were unconvinced that it was actually responsible for the pictures teams received. Of course, others were.

The first consensus-based team, as mentioned previously was undeniably the most negative and their language describe a tenuous relationship to the robot at best. Of all of the teams, it was the only one who at one point referred to the robot outside of the group, as "it." Upon the failure of the execution of a command set, the robot hit an obstacle and could no longer move the group discussion was as follows:

> *CT-1: It didn't even turn left*
> *PI: No because it can't*
> *MT-1: It went forward, left, forward*

It's unclear as to whether this is an indication that blame is being placed on the robot. If this is the case it oddly gives some volition to the robot but at the same time separates it from the team. Because the team acts on a consensus, placing blame on the robot would be one way of separating the mistake from the consensus itself, which had of course agreed upon the actions that led to this mistake.

"I would not use this rover" is an additional quote that came out of this consensus team, but this negativity was also shared with the other tightly coupled consensus team as well. That team, also utilizing the use of "we" in a way that seems to include the robot, does at one point say, "this machine sucks" lamenting its limitations in the field.

On the other conditional side, tightly coupled hierarchical teams, although they too questioned the robot's existence or functionality at times, did not demonstrate outwardly this negativity. In fact, these groups were eventually convinced of the robot's true existence because, as one team member states, "a human being could do this much faster." Again, this is recognition of the limitations of the robot but not necessarily a direct criticism as saying "it sucks" would be. In fact, both of these groups demonstrated a more obvious awareness of their robotic partner by making choices that considered the robot's size and spatial orientation in the experimental space. At one point on hierarchy team 2 a team member points out that a piece of tape demarcating the grid visible in the picture frame must be situated just below the camera of the robot.

To me this demonstrates a distinction the team made between the robot and its constituent parts. The robot has parts and each of these parts has a function. I believe that this again reflects what I observed earlier in that consensus-based group see the entirety of the robot as a tool while hierarchical ones, similar to how their team is divided into function, see the robot as a teammate with separated functions, but this theory will be discussed in the following chapter. Note also that hierarchy team 2 at one point described the terrain and obstacles ahead as “dangerous,” which is a curious choice of word unless it indicates at least some modicum of relationship or personification of their robotic teammate.

The loosely coupled groups developed the most conspicuous relationships with the S346 though. The loosely coupled consensus-based team was the only one to directly speak positively of their robotic teammate. Although at time recognizing its limitations, one instance where the team was reflecting on its success the conversation includes

> *MT-1: I think S346 has done a pretty good job so far*
> *MT-2: Definitely, he’s a trooper*

Not only is this a direct compliment of the robot’s capacity to complete the task, but also by individualizing the robot it implies that it has something to offer the team rather than just being used by it. Moreover, this excerpt is the only time a team gave their robotic teammate a pronoun, namely “he.” For this team the robot has taken on a masculine identity to which “he” can be referred.

The loosely coupled hierarchical team, although not featuring any notable moments in their dialogue that described their relationship to the robot did feature a unique method of operationalizing this connection. This group utilized a pencil to act as an “at home” proxy for the remote partner. The pencil was always oriented in whatever direction the robot was in its location so that it could be referred to with ease. This, unlike any other group, gave the robotic partner a physical presence within the group itself and perhaps suggests heightened interaction between human and robot teammates. For whatever reason, where social order was loosely defined, more creative and more varied instances of a relationship with the robot arose.

## 5. Discussion

Very few of my hypotheses when beginning this experiment found support in the data towards the end. While initially unfortunate, the revelation that they became nearly irrelevant given that they were predicated on a basic bipartite division of the data, when in reality a tripartite distinction emerged was truly exciting. By having a new and completely unexpected class of data, whose consistency held through all three types of analysis, I was able to reevaluate the ideas behind organizational structure and bring in a language of formal and informal processes that are essential to organizational sociology.

My first hypothesis was connected strongly to prior literature on task based team effectiveness. The notion that in general consensus teams find themselves more successful led me to believe that this would apply in this study as well. This was also affected by the idea that the Mars Exploration Rover team is considered a "successful" one; although of course there is no basis of comparison when discussing purely performance metrics given there have been so few rovers.

Of course, the data showed essentially the opposite. Under the bipartite distinction it became apparent that consensus-based groups were less successful at the goals of exploration and exploitation, finding fewer objects and exploring less space. However, it was impossible to deny that one consensus-based team was able to achieve comparable amounts of success in these areas.

This consensus group interestingly enough had many of the features I would have predicted to exist because of a consensus. They communicated efficiently, had an awareness of each other's roles, and showed evidence of mature shared mental models. However, given that these features were more present in the other hierarchy-based teams rather than consensus-based teams I found myself making observations that this group adopted pieces of the hierarchical power structure in order to find success. Which leads me now to say that the labels of "consensus" and "hierarchy" appear rather subjective given the context of a team-based experiment. Based on the task and what it demands, these two general conditions can be expressed in a variety of different ways and therefore a statement such as "frequent and efficient communication is a feature of consensus-based teams" also becomes irrelevant.

Instead of attempting to conclude as to which groups *are* in the objective sense what their condition labels them to be, it is more fruitful to discuss what strengths and weaknesses were gained by each team under the operationalization of the restrictions and condition which they were given. Therefore the labels of "tight" versus "loose", and "consensus" versus "hierarchy" are helpful as labels indicating the present similarities and differences between the emergent groups, but should not necessarily be read as implying objective qualities.

Gladstein et al. break down team effectiveness into three constituent parts being "group performance, satisfaction of group member needs, and the ability of the group to exist over time" (1984:500) These are also effective for determining where each of the three groups excelled at and where they fell short.

The first group of teams to discuss is the tightly coupled consensus-based teams. These teams were given the consensus condition that dictated they must find a unanimous consensus before moving forward. Ultimately, it were the observed qualitative features such as equal amounts of speaking between teammates (although interruptive), frequent sites of group decision-making, and numerous occasions of repetitions and clarification, that built towards the label of tightly coupled. There were no obvious hidden power dynamics on these teams. Regardless, performance-wise, these teams were not extremely successful. As compared to the other four teams, the teams in this cluster explored a notably smaller amount of space and discovered markedly fewer objects. However, these teams did excel at accuracy in that they made few mistakes in labeling the objects they did find.

I suggest that within the consensus condition, teams that fully embraced this value found themselves frequently hindered not only by the difficulty of achieving a unanimous vote, but also by the fact that their discussions led them towards more sites where consensus became necessary. These groups, as demonstrated by their transcripts, embraced consensus not only when crafting command sets, but also when identifying and locating obstacles and objects. Separate and numerous occasions of consensus and group deliberation occurred when discussing the end location of a given command set, which commands were needed to get there, how many pictures to take and where to take them, and what they had found before and after each execution of a command set. I believe that ultimately these two groups were forced to adopt a "slowest-ship-in-the-convoy" attitude towards command sets, shortening them to satisfy all team members. This would explain the resulting amount of space explored, which was fewer than 20% for both teams.

The error rate success can perhaps be explained by the same logic. Because uncertainty and disagreement were decidedly not an option for these two teams, objects and obstacles had to be placed with certainty. Moreover, here the map was a group resource, and although technically editable by anyone, was inevitably more restricted, as edits were also subjected to consensus. Because movement through the space depended on a detailed (though not necessarily accurate) map, and these groups chose not to make these details without unanimous certainty, they were also pushed to make command sets that worked towards the goal of achieving this certainty. Some of these command sets may have been limited in movement, overlapped already explored space, and generally taken precedence over the goal of *exploration*. Concerning shared mental models, especially important to perhaps the consensus requirements pushed these teams to adopt relatively simple models of the task that could be easily agreed upon and held by all, and less mature mental models can lead to lower productivity (Salas et al. 2005).

While team performance metrics were not particularly stellar, team member satisfaction and the possible longevity of teams in this group are a different matter. Although the way in which an individual valued himself was consistent across groupings, these two teams had relatively high measures of trust in their PI and other teammates as well. Especially in this condition, we would expect little difference in evaluations between the PI and the other team members given the flattened structure of the team. I would say that teams that stuck closely to the consensus are in fact particularly well suited to generating team member satisfaction at the least because that very measure is necessary for team action in many ways. In teams with such open dialogue and sharing of information, it would be difficult to conceal dissatisfaction, and thus those frequent moments of consensus gathering also serve as a touchstone for team member satisfaction.

It is also important to realize the inevitable inability to mirror the truly long-term teams behind the Mars Rover that this study took inspiration from. Although this study provided as many repeated opportunities for team decision making as possible, it is not a longitudinal study. However, even within the given 90 minutes, there is evidence of increased expertise of the team as a whole by the end, with later command sets gradually taking less time to complete. If consensus-based teams do in fact lead to increased member satisfaction and trust and these teams become more efficient as a result, it may suggest that over the long term, the poor initial performance metrics can be mitigated over time. Perhaps consensus-based teams are more subject to an “orientation-period,” that this study could not test for. I look to Tollinger (2006) here who has observed the consensus-based MER teams gradually decreasing the time spent making decisions as their time and experience as a team increased to give support to this speculation. Arguments for possible longevity could also be supported by the fact that these two teams also averaged the highest for ideal team size suggesting that, while perhaps not teams of 5, larger team sizes are sustainable under this condition.

How do the “tightly coupled” hierarchical teams compare in terms of strengths and weaknesses? It seems clear that the hierarchical condition, and fundamentally the ability to function without a consensus, allowed these teams to explore more of the map and discover more objects. By skipping the evidently time-consuming process of acquiring consensus, teams were able to submit larger command set for the S346. What is interesting is that consensus actually emerged as the more time consuming process in the first place. I expected the hierarchical decision making process to be the lengthier one, thinking about the time it would take for data to be organized, passed to one key spot or person, and synthesized and interpreted in order to come to a decision. What was unexpected was how the objective description of a hierarchy became task dependent. The collaborative nature of the task led to the hierarchy not necessarily being expressed as a power dynamic, but in the division of information and expertise. Data and information was hierarchicalized but not necessarily structures of power, so teammates would defer to the hierarchy defined by relevant expertise—which could reinterpret the hierarchy at each decision making point. Here we see again why the title of "tightly coupled" is more effective as just a label, as the collaborative nature of the task reinterpreted many of the facets of an objective hierarchy, if there is such a thing.

A weakness of avoiding consensus however was that error rates in object placement were markedly higher. Because these teams were able to operate, even comfortable, under uncertainties and approximations, they generally crafted command sets towards progress through new territory. As the map in this case was not a collaborative resource, the opinion of the movement technicians was often all the group had to work with and there was little need nor opportunity to debate this, or subject it to a collective decision. However, whatever mistakes were made then by a sub-team within the group, became a mistake for the whole group. Consensus, although time consuming, functioned as a successful corrective and confirmatory mechanism, and allowed teammates to exercise a type of "backup behavior" (Gao et al. 2014), that hierarchical teams did not utilize. Nor were hierarchy-based teams able to supplement this function with another, more aligned with their chosen operation. The loosely coupled hierarchical team managed to mitigate this by flattening access to data and allowing other members to demonstrate temporary expertise of data that was not necessarily "theirs."

As for team member satisfaction however, these teams seem just as, if not more satisfied with their PI and other teammates as "tightly coupled" consensus team members did. Although there is less evidence of these team members becoming more "used to" each other, as the decreasing time to complete command sets was not seen here, it could easily be that the process of hierarchical decision making had an inherent minimum time that was unavoidable. As I posited before, I believe this satisfaction and trust comes not from frequent interaction between equals, as it does within the consensus condition, but more from the separation of expertise that is a fundamental character of the designed condition, and the resulting success. As team members on these teams found that they were successfully exploring and exploiting their space simply by trusting and utilizing the knowledge of their teammates, they became more comfortable repeating this process; a sort of positive feedback loop. The results of the exit survey support the notion that when the division of responsibility is clear within the social order, trust and satisfaction increase. From here it is interesting to speculate that because team members know where responsibility lies, they are also inherently able to place blame. Do these team members then have to work to establish themselves as trustworthy to their peers through repeated success? What would happen should they personally fail?

I also lean towards the conclusion that the "tightly coupled" hierarchical teams have the ability to exist over time like the consensus-based ones, but that they will not see the same improvement. The system these teams work with is error prone but provides little opportunity for correction based on my observations. This ultimately raises the important question of not which team is more effective, but more effective in what context. MacMillan et al. (2004) stress the relevancy of context and task requirements when crafting ideal teams, and in my study part of this context is whether the task is long or short term.

Finally though, given that the task assigned to the teams was to maximize exploration and exploitation of the experimental space, and the scripts placed heavy emphasis on the notion of efficiency, I am compelled to say that the most successful teams were the loosely coupled ones. For the amount of space of explored, which was on the higher end for both, they discovered 60-70% of all of the objects, a number not exceeded by any other team in any other condition. Moreover, these teams managed to make few mistakes when placing these objects on the map, if any at all. It appears as though the performance strengths of the hierarchical condition and the consensus condition were adopted within these teams to complete the task objectively well.

Both of these teams managed to find a way of operating that allowed them to share resources collaboratively, while concurrently establishing, or maintaining, areas of expertise within the larger team. This ultimately allowed team members to truly work together during discussion sessions but shorten the amount of time it would take other groups to parse through information. There are also many examples on both of these teams of teammates being able to work separately as well, with perhaps two members working with one piece of data and two to three working on another, coming together at the end to either explain the decision or insure that a consensus is still being achieved. For Salas et al. (2004), Mickan & Rodger (2000), and the others who study teamwork, this would be evidence of effective shared mental models for these teams demonstrate a shared understanding of their goals, task, and how they work together in this specific context. Interestingly though, these models may not include a shared understanding of each other in terms of roles and value added. On these teams the maintenance or establishment of some divisions, alongside the breakdown of others, evidently led to ambiguity in roles and team function that effected trust and satisfaction among team members. Teammates of the loosely coupled teams trusted their leaders and teammates less than the other groups while also feeling as though they too were not very trusted by their teammates. The ability to pick apart aspects of team roles and move them to whatever was convenient at a given time was successful for short-term success, but possibly detrimental to team longevity. This shows that not only the nature of task, but also the time frame in which it is expected to be completed, can dictate which team structure is most effective at a given time. Like MacMillan et al. (2004), Singh et al. (2011) also understand the importance of task, and understand the possible tradeoff between team building and actual productivity, with their study showing that non-flat teams were best for short-term productivity.

In my study, flexibility and the ability to adopt informal methods of achieving consensus, include teammates, explain strategies, and implicitly establish new power dynamics, were very helpful in the field for a team consisting of untrained participants with little time to adjust, but this might not be the case for other types of collaborative work and other types of participants.

*The Robotic Team Member*

The robot itself is an interesting component of this experiment because its role on the team seems to stand apart from previous studies barring the Mars Exploration Rovers themselves. Because of the camera's remote location, the S346 inevitably becomes a proxy for sight and presence. By being involved in a collaborative task, it by design becomes included into the "we" that makes up the team. This is ultimately a direct parallel to Vertesi's findings about *Spirit and* Opportunity in which the "rover activities and rover agency are predicated on [the] internal organization, expressed through the robot's body" (Vertesi 2015:179). Although it is the robot that is physically "facing this way" and the robot that "is looking at that," because its actions are inherently a product of team collaboration or perhaps even a member of that collaboration, being part of the inclusive speech is not surprising. "We" is an essential linguistic element throughout this study, and what is further interesting was that more than just a collectivist type of organization could be expressed through the body of the S346. Both a "hierarchy" and a "consensus" were ultimately expressed at some level of success.

Another complication in the basis of comparison to prior research is that that research, even the experiments that inspired this study's procedure, tend to utilize autonomous or semi-autonomous robots. In these cases humans work alongside robotic partners and I believe that this ultimately allows the human component to separate itself from the robotic one, and research shows that oftentimes humans still would prefer other human teammates (Gombolay et al. 2014). I believe that in my study the presence of the robot catalyzed the collaborative nature of the task because regardless of condition, information from the S346 was expressed as "we" even when expertise was unbalanced and social order unclear or even tenuous.

Additionally, although the robot cannot succeed without the team behind it, the team cannot function without the robot either. By pushing the robot into the "we" space, I believe the team is forced to consider the limitations and abilities of the robot during the experiment, and this too affects the factors comprising team effectiveness. This is clearly evident as well by the frequent acknowledgement of the admittedly limited robot with which they were working. Had the S346 been more advanced and the task more complicated, the relationship with the robot could have been expressed even more observably. Vertesi (2016) speaks volumes of the MER teams' distress over the possibility of rover death, and the constant adaptation the entire team makes when the rover's physical status and ability to perform its role changes.

Now the effect of the teams' relationship with the S346 on the success of the task, and vice versa, remains partly unclear. However, from this study it is undeniable that a positive or more conspicuous relationship with the robot was present on the most successful teams, being the loosely coupled ones, at least in terms of performance. A positive emotional relationship to the robotic teammate and "embodying" the robot's physiology, examples of each which were present on the loosely coupled teams in this study, has been implicated by Vertesi (2016) to be influential to the success of the MER teams. However in those cases the collectivist and consensus-based structure seems to be responsible for facilitating these factors. I believe that in my study, given that strictly following the consensus structure diverted too many resources towards discussion over details and identification—activities the robot inherently could not partake in, or as readily invoked during—it weakened the relationship to the S346. The more loosely defined social structures managed to more often discuss matters of progression through the experiment, and although their relationships to each other appeared unclear at time, the robot was an essential focal point of this team and task. Additionally these teams, with the S346, were able to produce images that were both enlightening and proof of a successful strategy. This reinforced and validated the methodology of the team operation and how the relationship to the robot was being formed and expressed, similar to the function of images throughout Vertesi's (2015, 2016) work.

## 6. Conclusion

The rovers of the Mars Exploration Rover mission are highly successful. The teams that they are a part of have a unique relationship to these rovers, and act within highly flattened team structures in order to produce this success. My study intended to question the team structure element in this equation and observe how its variance affected the productivity of a collaborative team working with a remote robot. I also wanted to observe the development of their relationships to their robotic teammate and how this would vary as well.

In this study I created teams of students to work with each other and a remote robotic “rover” teammate to explore an experimental space of obstacles as widely as possible while finding objects of interest. Essentially the task was a physical manifestation of previous search and rescue tasks with larger teams of five. Within this experiment two conditions existed that dictated the team structure or organizational form the team was meant to follow, consensus or hierarchy. Scripts described teammates respective roles and goals within their teams accordingly. Additionally, seating arrangement was varied between conditions to better influence the flow of data in a way that reflected the given condition, either more laterally or vertically. Teams were then given 90 minutes to repeatedly make decisions about “command sets” their robotic teammate would initiate that described both the movement of the robot as well as pictures it would take. The teams were given a max of 10 minutes per command set and these commands were executed in bulk to reflect, although on a severely smaller scale, the day to day repeated discussion sessions that the MER teams have to decide how their rover will act 400 million kilometers away.

What I found was that instead of differences being divided in half along conditional lines, some teams found room to reinterpret their given restrictions, thus giving rise to not two, but three clusters within the data. The emergence of these three groups, the tightly coupled hierarchies, the tightly coupled consensus groups, and the loosely coupled version of each, were further supported both by qualitative observations of their discussions and exit survey data.

Tightly coupled consensus-based teams explored the least and discovered the least but made few errors in the process. It seemed as though achieving consensus itself was an arduous process and that consensus had to be achieved very often throughout their 90 minutes. Multiple voices led to certainty, but only of the lowest common denominator, leading to slowed progression and even backtracking.

Tightly coupled hierarchical-based teams allowed their sub-teams to operate more independently and teammates accepted the implied expertise of one another. Thus lengthy discussions were minimized and progression was emphasized, but with the acceptance of expert opinion came the acceptance of their mistakes as well. These teams explored more and discovered more but featured more mistakes.

Loosely coupled teams emerged as the most successful in terms of productivity exploring the most space and finding the most objects with the fewest errors. The loosely coupled hierarchical team was able to flatten access to data allowing for multiple team members to check each other and exercise more backup behavior. The features of a hierarchical team described above were present, bringing with them their advantages, and adoption of some “consensus-style” features, or rather aspects present on the teams in the opposite condition, led to this success.

The loosely coupled consensus team was able to more firmly distinguish between roles within the team and leaned towards recognition of implicit expertise. The act of achieving consensus was minimized or informalized as often team members could participate in the consensus merely through linguistic choices without being fully invested. In this way they limited the disadvantages consensus and could orient themselves toward progress.

However, given that team effectiveness is not only a factor of success but also of satisfaction, loosely coupled teams acting within informal structures featured lower levels of trust and value on their teams likely because the social order was unclear. It was at this point though that I still concluded that for the given *task*, 90 minutes of exploration and exploitation, these teams were the most successful. They were the best suited towards its successful completion. I also realized the successes and limitations, or opportunities for further research, in this study.

If my original research aim was to explore the effects of team structure on decision-making and success, as well as look at how the robot mediates or is affected by the condition, I feel obligated to refine it now by adding that the task was a short-term collaborative task that required inexpert teams to adjust to their working conditions and each other in limited time. And in this scenario, teams that felt they could operate outside the formal structure of their condition were more successful.

"Task" had further reaching effects than originally anticipated as 90 minutes and inexpert teams significantly affected how teams operated and what was important for success. But the study revealed much concerning the development of teamwork on short-term collaborative tasks

Ultimately, clear social order promoted team longevity and membership trust, but the adjustment period to this social order consumed time and resources that would have been helpful to these teams within the 90 minutes they had to complete their task. The loosely coupled teams, which were able to customize their team organization outside of the formal restrictions, and even change it on the fly, were better suited to continually be focused on the task, even though they were not focused on team organization. In these scenarios, operating within informal structures required something, I argue the robot, to maintain team structure where the formal structures could not.

Where the constraints of the task changed the way in which a consensus and hierarchy could be expressed and whether they would be successful, it did not change the importance of a human-robot relationship. Like the relationships present between MER scientists and *Spirit* and *Opportunity*, the relationship to the S346 was central to the maintenance of social order; but where the MER teams' relationship seems to be predicated on consensus-based team structure, the teams' relationships in my study were not. These relationships also lacked, or could not actively demonstrate in 90 minutes, the multifaceted depth of the MER ones which appear paramount to long-term success.

Any positive relationship with the robot, however temporary, was more important than long-term social structure, and this could be achieved within a variety of organizational conditions. However it is unclear whether this relationship could be maintained over the long-term without the consensus-based structure that has shown to be at least partly responsible for the success of the Mars Rovers.

Further research extends beyond many additional trials to solidify these results. There are a variety of changes to the experiment that could have observable effects on the experiment. Early on I thought about how varying and further specifying the tasks of sub-teams members would affect the decision making process. One such variation was to apply the titles of "cartographer," instead of "movement technician," and "geologist," instead of camera technician. Cartographers would be tasked with accurately drawing a map of the experimental space, instead of just exploring it physically, and geologists would need to take pictures of all of the objects within one space of it instead of only close enough to identify. I wondered if this would further separate the sub-teams goals, increase conflict, and weaken solidarity, and ultimately how this would effect which condition would prove more successful.

I also believe that increasing the time of each trial would have significant effects on the results given how the restriction of time severely altered the nature of the task as compared to its inspiration. With more time I'd like to observe at which point a consensus "matures" and complete is orientation and if any additional advantages are gained at this point. More time would also allow for the possibility of more mistakes. This would be especially relevant in relation to my speculation that on hierarchical teams, team members have to continually prove their success and that blame placement is more obvious. With more time, these team members have more opportunity to falter and how this affects how their teammates trust them, what it means to be "expert," and where blame is placed could be observed.

In this study, the goals, the time period, the risks, the stakes, and the level of collaboration, all intimately worked together to produce finely tuned scenarios that called for finely tuned teams. Combine this with the general notion that robots are making tasks easier for all of us, and it results in robots becoming integral parts of these finely tuned teams. However, we cannot force the human component to work successfully alongside the robotic. In fact, even if people prefer to work with other people it is difficult still, to have them work together as well. What we can do is work to identify the advantages and disadvantages of team organizations, in this case ones involving a remote robotic partner, and the type of task they are perhaps best suited to accomplish, closing gaps as small as a surgical incision and as large as the distance between two planets.

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# 8. Appendices

## Appendix A. Consensus Condition Scripts – Principal Investigator, Camera Technician, Movement Technician

You are employees for WallaceTech, a leading Robotics Company specializing in small research robots. You all work together on a team designing the "S346." Unfortunately the deadline for a functional robot is approaching and your team is almost over budget in development and testing costs. Your superiors are not happy with the proposed S346 and have proposed a field test to demonstrate the S346's viability. This field test will be your team and your robot's final chance to prove it was worth the investment or the project will be canceled.

S346 performs as follows. Being a simple research robot it can handle simple commands such as forward, backward, and turning in either direction. The S346 takes a series of commands in bulk and performs them remotely. Each "forward" or "backward" command correlates to 1 foot of movement. Each "turn" command correlates to a 90-degree rotation in the specified direction. The robot can take pictures and will do so with the command "picture." The camera can be tilted 45 degrees up or down with the command "camera up" or "camera down"

As the S346 is designed for remote research applications, often those in which humans would be to large to enter the research space, the superiors have developed the field test in a way they believe to be an appropriate assessment. The S346 will be placed in a room filled with obstacles and more importantly, objects, which your team must locate. Your research team will be placed in a separate room and not have access to the S346 directly. Instead you will utilize the picture command to assess where the S346 is at any given time.

As is typical of teams here at WallaceTech, they are comprised of 5 members. There is a Principal Investigator, 2 camera technicians, and 2 movement technicians. However, as we have tested that open dialogue and collaboration is the most successful for robot teams, all team members must come to a unanimous consensus for any given set of actions.

The field test will be run as follows. The bosses at WallaceTech want to see the maximization of **exploration** and **exploitation**. They want to see that the S346 and the team *can locate every object and explore the entirety of the space efficiently*. Thus your team has 90 minutes to complete the task and can choose to end before time is up if they believe the task complete.

The team will begin with a picture from the Robot's point of view in the starting location and an empty map of the space, which they are recommended to fill in as appropriate. From here, your team will have 10 minutes to decide on a command set to send to the S346. *Remember that all 5 members of the group must agree on each course of action*. Any pictures taken will be available at the end of the initiated command set and will be viewable on a separate device. The team will then have another 10 minutes to decide upon another command set and this cycle will repeat until time is up or the team deems the test completed.

*Scripts differ for participants at this point depending on assigned role*

*Principal Investigator*

You, **as the Principal Investigator**, should learn from your teammates the limitation of the S346 and try to coordinate the desires of the camera and movement technician teams. You should gather necessary information and facilitate the success of your team. As consensus is essential, you may often find yourself in an essential role in achieving agreement between your teammates. Thus, you must find a way to effectively balance the diverse challenges of the S346 namely locating every object accurately **and** exploring/understanding the whole testing space quickly and efficiently.

*Camera Technician*

You**, as the camera technician**, should know that in a real life application of the S346, memory is limited, the more pictures taken the slower transfer will be. Thus limiting the number of pictures is paramount to smooth performance. The camera functionality of the S346 is extremely important for demonstrating the value of your job and thus keep in mind your goals of locating and taking pictures every object

*Movement Technician*

You, **as the movement technician**, should know that in a real life application of the S346, time is limited thus the speed and efficiency of exploration are paramount to successful performance. You know that to demonstrate the value of you job, the S346 must move effectively and quickly throughout the space, exploring as much as possible in as little time as possible, gathering a broad understanding of the *whole* environment. The more space explored in less time, the better, and these will be factored into your efficiency scores.

*Scripts return to being identical*

Good luck, the future of the S346 is in your hands. If you have any further questions you may direct them to the experiment coordinator assigned by the WallaceTech leadership

Rules for whole team:

- Obstacles are not constrained to single boxes on the grid and may be placed at intersections between grid boxes
    - Any space not completely filled by an obstacle is accessible to the S346
- Please mark down on the map where and at what time objects were located

On each command set write what number command set it is and the time it took to complete it (1-10 minutes)

## Appendix B. Hierarchy Condition Scripts – Principal Investigator, Camera Technician, Movement Technician

You are employees for WallaceTech, a leading Robotics Company specializing in small research robots. You all work together on a team designing the "S346." Unfortunately the deadline for a functional robot is approaching and your team is almost over budget in development and testing costs. Your superiors are not happy with the proposed S346 and have proposed a field test to demonstrate the S346's viability. This field test will be your team and your robot's final chance to prove it was worth the investment or the project will be canceled.

S346 performs as follows. Being a simple research robot it can handle simple commands such as forward, backward, and turning in either direction. The S346 takes a series of commands in bulk and performs them remotely. Each "forward" or "backward" command correlates to 1 foot of movement. Each "turn" command correlates to a 90-degree rotation in the specified direction. The robot can take pictures and will do so with the command "picture." The camera can be tilted 45 degrees up or down with the command "camera up" or "camera down"

As the S346 is designed for remote research applications, often those in which humans would be to large to enter the research space, the superiors have developed the field test in a way they believe to be an appropriate assessment. The S346 will be placed in a room filled with obstacles and more importantly, objects, which your team must locate. Your research team will be placed in a separate room and not have access to the S346 directly. Instead you will utilize the picture command to assess where the S346 is at any given time.

As is typical of teams here at WallaceTech, they are comprised of 5 members, the Principal Investigator, two camera technicians, and two movement technicians. The Principal Investigator is the ultimate decision making authority on command sets for the S346 and must balance the concerns of the sub-teams under his/her control. Camera technicians should voice their information about camera concerns and decisions to the PI. Movement technicians should likewise pass along their movement recommendations to the PI. Information and opinions can be discussed between team members but the PI will be responsible for formulating a plan to pass to the S346.

The field test will be run as follows. The bosses at WallaceTech want to see the maximization of **exploration** and **exploitation**. They want to see that the S346 and the team can *locate every object and explore the entirety of the space quickly and efficiently*. Thus your team has 90 minutes to complete the task and can choose to end before time is up if they believe the task complete.

The camera tech team will begin with a picture from the Robot's point of view in the starting location and the movement tech team will have an empty map of the space, which they are recommended to fill in as appropriate. From here, your team will have 10 minutes to decide on a command set to send to the S346. *Remember that the Principal Investigator has the deciding authority and any plans must be formulated and passed to the S346 through him/her.* Any pictures taken will be available initially to the camera tech team at the end of the initiated command set and will be viewable on a separate device. The team will then have another 10 minutes to decide upon another command set and this cycle will repeat until time is up or the team deems the test completed.

*Scripts differ for participants at this point depending on assigned role*

*Principal Investigator*

You**, as the Principal Investigator**, have the ultimate authority on your team and thus are responsible for both listening to (and sometimes ignoring) the opinions of your teammates. It is up to you to gather the information you need from your sub-teams to complete your task. You must find a way to effectively balance the diverse challenges of the S346 namely locating every object accurately **and** exploring/understanding the whole testing space quickly and efficiently.

*Camera Technician*

You**, as the camera technician**, will be placed at the monitor with initial access to any pictures that arrive. You can share this information upon request and should know that in a real life application of the S346, memory is limited. The more pictures taken the slower transfer will be. Thus limiting the number of pictures is paramount to smooth performance. The camera functionality of the S346 is extremely important for demonstrating the value of your job and thus you must make clear to the Principal Investigator your goals of **locating and taking pictures every object**.

*Movement Technician*

You, **as the movement technician**, will have initial access, editing power, and control of the map, and can share this information upon request. You know that to demonstrate the value of you job the S346 must move effectively and quickly throughout the space, exploring as much as possible in as little time as possible**.** Speed and efficiency of exploration are paramount to successful performance in the field and a broad understanding of the testing space is sometimes more important than minute details or specific locations. The more space explored in less time, the better, and these will be factored into your efficiency scores and thus make sure the Principal Investigator is aware of these goals and concerns.

*Scripts return to being identical*

Good luck, the future of the S346 is in your hands. If you have any further questions you may direct them to the experiment coordinator assigned by the WallaceTech leadership.

Rules for whole team:

- Obstacles are not constrained to single boxes on the grid and may be placed at intersections between grid boxes
    - Any space not completely filled by an obstacle is accessible to the S346
- Please mark down on the map where and at what time objects were located
- On each command set write what number command set it is and the time it took to complete it (1-10 minutes)

## Appendix C. Exit Survey

Post-Experimental Survey

Trial Number (See Instructor)____________

Experimental Conditions (See Instructor)_____________

Role On Team (Circle One)

Principal Investigator    Camera Technician    Movement Technician

In each 10 minute planning period, on average how valuable would you rate your contribution? (1 = never considered, 7 = absolutely essential)

1 2 3 4 5 6 7

In each 10 minute planning period, on average how valuable would your teammates rate your contribution? (1 = never considered, 7 = absolutely essential)

1 2 3 4 5 6 7

How much do you trust the Principal Investigator? (1= not at all, 7 = completely)

1 2 3 4 5 6 7

How much did you trust the rest of your teammates? (1= not at all, 7 – completely)

1 2 3 4 5 6 7

How long on average do you believe each planning session took? (in minutes)

1 2 3 4 5 6 7 8 9 10

What is the ideal number of people for a team in control of a robot?

1 2 3 4 5 >5

How limited was the robot in its capabilities towards task completion? (1= excessively limited, 7= Highly flexible)

1 2 3 4 5 6 7

How well did your team adjust to the limitations and abilities of the robot? (1= not well adjusted, 7= very well adjusted)

1 2 3 4 5 6 7

What was more essential to the success of the task and of the team? (Circle One)

More Pictures More Exploration

Would you encourage WallaceTech to continue funding for the S346? (1= stop funding immediately, 7= continue and increase funding)

1 2 3 4 5 6 7

**Appendix D. Survey Data: By Condition**

| Trial | Condition | Role | Q1 | Q2 | Q3 | Q4 | Q5 | Over/Under Estimate? | Q6 | Q7 | Q8 | Q9 | Q10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | PI | 4 | 6 | 7 | 7 | 4 | Over | 4 | 4 | 7 | Exp | 5 |
| 1 | C | MT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 3 | 5 | Exp | 4 |
| 1 | C | MT | 5 | 4 | 4 | 6 | 3 | Over | 3 | 3 | 6 | Pic | 3 |
| 1 | C | CT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 2 | 5 | Pic | 3 |
| 1 | C | CT | 4 | 5 | 7 | 7 | 5 | Over | 4 | 4 | 6 | Exp | 4 |
| 2 | C | PI | 3 | 3 | 7 | 5 | 5 | Over | 2 | 2 | 5 | Exp | 2 |
| 2 | C | MT | 5 | 5 | 3 | 5 | 4 | Over | 3 | 2 | 5 | Exp | 5 |
| 2 | C | MT | 5 | 4 | 3 | 3 | 4 | Over | 2 | 2 | 5 | Pic | 4 |
| 2 | C | CT | 2 | 2 | 5 | 6 | 4 | Over | 3 | 5 | 6 | Pic | 4 |
| 2 | C | CT | 7 | 7 | 1 | 3 | 5 | Over | 3 | 6 | 7 | Pic | 7 |
| 3 | C | PI | 4 | 4 | 7 | 7 | 5 | Over | 5 | 4 | 6 | Exp | 4 |
| 3 | C | MT | 4 | 4 | 5 | 5 | 4 | Accurate | 3 | 3 | 6 | Pic | 5 |
| 3 | C | MT | 5 | 4 | 4 | 5 | 5 | Over | 4 | 3 | 5 | Exp | 3 |
| 3 | C | CT | 6 | 6 | 5 | 5 | 4 | Accurate | 3 | 5 | 6 | Exp | 6 |
| 3 | C | CT | 3 | 3 | 7 | 7 | 4 | Accurate | 3 | 3 | 6 | Exp | 5 |
| | | **Average** | **4.4667** | **4.4667** | **5.1333** | **5.5333** | **4.1333** | | **3.2000** | **3.4000** | **5.7333** | | **4.2667** |
| 1 | H | PI | 5 | 4 | 7 | 7 | 3 | Accurate | 4 | 3 | 7 | Pic | 1 |
| 1 | H | MT | 6 | 5 | 6 | 4 | 5 | Over | 1 | 4 | 6 | Exp | 4 |
| 1 | H | MT | 4 | 4 | 3 | 5 | 3 | Accurate | 4 | 3 | 6 | Pic | 2 |
| 1 | H | CT | 6 | 6 | 6 | 6 | 4 | Accurate | 6 | 5 | 7 | Pic | 7 |
| 1 | H | CT | 3 | 6 | 6 | 6 | 5 | Over | 3 | 6 | 6 | Exp | 6 |
| 2 | H | PI | 2 | 1 | 1 | 3 | 5 | Accurate | 3 | 2 | 3 | Pic | 1 |
| 2 | H | MT | 4 | 6 | 6 | 7 | 6 | Over | 2 | 3 | 5 | Exp | 4 |
| 2 | H | MT | 5 | 5 | 7 | 7 | 6 | Over | 4 | 4 | 5 | Exp | 6 |
| 2 | H | CT | 5 | 6 | 4 | 7 | 6 | Over | 4 | 4 | 6 | Pic | 6 |
| 2 | H | CT | 6 | 6 | 5 | 5 | 6 | Over | 2 | 4 | 7 | Pic | 6 |
| 3 | H | PI | 4 | 3 | 5 | 6 | 5 | Over | 3 | 3 | 6 | Exp | 5 |
| 3 | H | MT | 5 | 4 | 5 | 5 | 4 | Accurate | 1 | 2 | 5 | Exp | 7 |
| 3 | H | MT | 3 | 2 | 6 | 6 | 3 | Under | 3 | 4 | 5 | Exp | 5 |
| 3 | H | CT | 7 | 6 | 6 | 3 | 6 | Over | 3 | 4 | 3 | Pic | 4 |
| 3 | H | CT | 3 | 3 | 7 | 7 | 7 | Over | 3 | 5 | 6 | Pic | 3 |
| | | **Average** | **4.5333** | **4.4667** | **5.3333** | **5.6000** | **4.9333** | | **3.0667** | **3.7333** | **5.5333** | | **4.4667** |

**Appendix E. Survey Data: By Condition and Sub-team/PI**

| Trial | Condition | Role | Q1 | Q2 | Q | Q4 | Q5 | Over/Under | Q6 | Q7 | Q8 | Q9 | Q10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | PI | 4 | 6 | 7 | 7 | 4 | Over | 4 | 4 | 7 | Exp | 5 |
| 2 | C | PI | 3 | 3 | 7 | 5 | 5 | Over | 2 | 2 | 5 | Exp | 2 |
| 3 | C | PI | 4 | 4 | 7 | 7 | 5 | Over | 5 | 4 | 6 | Exp | 4 |
| | | **Average** | **3.6667** | **4.3333** | **7.0000** | **6.3333** | **4.6667** | | **3.6667** | **3.3333** | **6.0000** | | **3.6667** |
| | | | | | | | | | | | | | |
| 1 | H | PI | 5 | 4 | 7 | 7 | 3 | Accurate | 4 | 3 | 7 | Pic | 1 |
| 2 | H | PI | 2 | 1 | 1 | 3 | 5 | Accurate | 3 | 2 | 3 | Pic | 1 |
| 3 | H | PI | 4 | 3 | 5 | 6 | 5 | Over | 3 | 3 | 6 | Exp | 5 |
| | | **Average** | **3.6667** | **2.6667** | **4.3333** | **5.3333** | **4.3333** | | **3.3333** | **2.6667** | **5.3333** | | **2.3333** |
| | | | | | | | | | | | | | |
| 2 | C | CT | 2 | 2 | 5 | 6 | 4 | Over | 3 | 5 | 6 | Pic | 4 |
| 2 | C | CT | 7 | 7 | 1 | 3 | 5 | Over | 3 | 6 | 7 | Pic | 7 |
| 3 | C | CT | 6 | 6 | 5 | 5 | 4 | Accurate | 3 | 5 | 6 | Exp | 6 |
| 3 | C | CT | 3 | 3 | 7 | 7 | 4 | Accurate | 3 | 3 | 6 | Exp | 5 |
| 1 | C | CT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 2 | 5 | Pic | 3 |
| 1 | C | CT | 4 | 5 | 7 | 7 | 5 | Over | 4 | 4 | 6 | Exp | 4 |
| 1 | C | MT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 3 | 5 | Exp | 4 |
| 1 | C | MT | 5 | 4 | 4 | 6 | 3 | Over | 3 | 3 | 6 | Pic | 3 |
| 2 | C | MT | 5 | 5 | 3 | 5 | 4 | Over | 3 | 2 | 5 | Exp | 5 |
| 2 | C | MT | 5 | 4 | 3 | 3 | 4 | Over | 2 | 2 | 5 | Pic | 4 |
| 3 | C | MT | 4 | 4 | 5 | 5 | 4 | Accurate | 3 | 3 | 6 | Pic | 5 |
| 3 | C | MT | 5 | 4 | 4 | 5 | 5 | Over | 4 | 3 | 5 | Exp | 3 |
| | | **Average** | **4.6667** | **4.5000** | **4.6667** | **5.3333** | **4.0000** | | **3.0833** | **3.4167** | **5.6667** | | **4.4167** |
| 1 | H | MT | 6 | 5 | 6 | 4 | 5 | Over | 1 | 4 | 6 | Exp | 4 |
| 1 | H | MT | 4 | 4 | 3 | 5 | 3 | Accurate | 4 | 3 | 6 | Pic | 2 |
| 2 | H | MT | 4 | 6 | 6 | 7 | 6 | Over | 2 | 3 | 5 | Exp | 4 |
| 2 | H | MT | 5 | 5 | 7 | 7 | 6 | Over | 4 | 4 | 5 | Exp | 6 |
| 3 | H | MT | 5 | 4 | 5 | 5 | 4 | Accurate | 1 | 2 | 5 | Exp | 7 |
| 3 | H | MT | 3 | 2 | 6 | 6 | 3 | Under | 3 | 4 | 5 | Exp | 5 |
| 1 | H | CT | 6 | 6 | 6 | 6 | 4 | Accurate | 6 | 5 | 7 | Pic | 7 |
| 1 | H | CT | 3 | 6 | 6 | 6 | 5 | Over | 3 | 6 | 6 | Exp | 6 |
| 2 | H | CT | 5 | 6 | 4 | 7 | 6 | Over | 4 | 4 | 6 | Pic | 6 |
| 2 | H | CT | 6 | 6 | 5 | 5 | 6 | Over | 2 | 4 | 7 | Pic | 6 |
| 3 | H | CT | 7 | 6 | 6 | 3 | 6 | Over | 3 | 4 | 3 | Pic | 4 |
| 3 | H | CT | 3 | 3 | 7 | 7 | 7 | Over | 3 | 5 | 6 | Pic | 3 |
| | | **Average** | **4.7500** | **4.9167** | **5.5833** | **5.6667** | **5.0833** | | **3.0000** | **4.0000** | **5.5833** | | **5.0000** |

**Appendix F. Survey Data: By Condition and Role**

| Trial | Condition | Role | Q1 | Q2 | Q3 | Q4 | Q5 | Over/Under | Q6 | Q7 | Q8 | Q9 | Q10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | PI | 4 | 6 | 7 | 7 | 4 | Over | 4 | 4 | 7 | Exp | 5 |
| 2 | C | PI | 3 | 3 | 7 | 5 | 5 | Over | 2 | 2 | 5 | Exp | 2 |
| 3 | C | PI | 4 | 4 | 7 | 7 | 5 | Over | 5 | 4 | 6 | Exp | 4 |
| | | **Average** | **3.6667** | **4.3333** | **7.0000** | **6.3333** | **4.66667** | | **3.6667** | **3.3333** | **6.0000** | | **3.6667** |
| 1 | H | PI | 5 | 4 | 7 | 7 | 3 | Accurate | 4 | 3 | 7 | Pic | 1 |
| 2 | H | PI | 2 | 1 | 1 | 3 | 5 | Accurate | 3 | 2 | 3 | Pic | 1 |
| 3 | H | PI | 4 | 3 | 5 | 6 | 5 | Over | 3 | 3 | 6 | Exp | 5 |
| | | **Average** | **3.6667** | **2.6667** | **4.3333** | **5.3333** | **4.3333** | | **3.3333** | **2.6667** | **5.3333** | | **2.3333** |
| | | | | | | | | | | | | | |
| 2 | C | CT | 2 | 2 | 5 | 6 | 4 | Over | 3 | 5 | 6 | Pic | 4 |
| 2 | C | CT | 7 | 7 | 1 | 3 | 5 | Over | 3 | 6 | 7 | Pic | 7 |
| 3 | C | CT | 6 | 6 | 5 | 5 | 4 | Accurate | 3 | 5 | 6 | Exp | 6 |
| 3 | C | CT | 3 | 3 | 7 | 7 | 4 | Accurate | 3 | 3 | 6 | Exp | 5 |
| 1 | C | CT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 2 | 5 | Pic | 3 |
| 1 | C | CT | 4 | 5 | 7 | 7 | 5 | Over | 4 | 4 | 6 | Exp | 4 |
| | | **Average** | **4.5000** | **4.6667** | **5.1667** | **5.6667** | **4.1667** | | **3.1667** | **4.1667** | **6.0000** | | **4.8333** |
| 1 | H | CT | 6 | 6 | 6 | 6 | 4 | Accurate | 6 | 5 | 7 | Pic | 7 |
| 1 | H | CT | 3 | 6 | 6 | 6 | 5 | Over | 3 | 6 | 6 | Exp | 6 |
| 2 | H | CT | 5 | 6 | 4 | 7 | 6 | Over | 4 | 4 | 6 | Pic | 6 |
| 2 | H | CT | 6 | 6 | 5 | 5 | 6 | Over | 2 | 4 | 7 | Pic | 6 |
| 3 | H | CT | 7 | 6 | 6 | 3 | 6 | Over | 3 | 4 | 3 | Pic | 4 |
| 3 | H | CT | 3 | 3 | 7 | 7 | 7 | Over | 3 | 5 | 6 | Pic | 3 |
| | | **Average** | **5.0000** | **5.5000** | **5.6667** | **5.6667** | **5.6667** | | **3.5000** | **4.6667** | **5.8333** | | **5.3333** |
| | | | | | | | | | | | | | |
| 1 | C | MT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 3 | 5 | Exp | 4 |
| 1 | C | MT | 5 | 4 | 4 | 6 | 3 | Over | 3 | 3 | 6 | Pic | 3 |
| 2 | C | MT | 5 | 5 | 3 | 5 | 4 | Over | 3 | 2 | 5 | Exp | 5 |
| 2 | C | MT | 5 | 4 | 3 | 3 | 4 | Over | 2 | 2 | 5 | Pic | 4 |
| 3 | C | MT | 4 | 4 | 5 | 5 | 4 | Accurate | 3 | 3 | 6 | Pic | 5 |
| 3 | C | MT | 5 | 4 | 4 | 5 | 5 | Over | 4 | 3 | 5 | Exp | 3 |
| | | **Average** | **4.8333** | **4.3333** | **4.1667** | **5** | **3.8333** | | **3.0000** | **2.6667** | **5.3333** | | **4** |
| 1 | H | MT | 6 | 5 | 6 | 4 | 5 | Over | 1 | 4 | 6 | Exp | 4 |
| 1 | H | MT | 4 | 4 | 3 | 5 | 3 | Accurate | 4 | 3 | 6 | Pic | 2 |
| 2 | H | MT | 4 | 6 | 6 | 7 | 6 | Over | 2 | 3 | 5 | Exp | 4 |
| 2 | H | MT | 5 | 5 | 7 | 7 | 6 | Over | 4 | 4 | 5 | Exp | 6 |
| 3 | H | MT | 5 | 4 | 5 | 5 | 4 | Accurate | 1 | 2 | 5 | Exp | 7 |
| 3 | H | MT | 3 | 2 | 6 | 6 | 3 | Under | 3 | 4 | 5 | Exp | 5 |
| | | **Average** | **4.5000** | **4.3333** | **5.5000** | **5.6667** | **4.5000** | | **2.5000** | **3.3333** | **5.3333** | | **4.6667** |

**Appendix G. Survey Data: By Revealed Grouping**

| Trial | Condition | Role | Q1 | Q2 | Q3 | Q4 | Q5 | Over/Under Estimate? | Q6 | Q7 | Q8 | Q9 | Q10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | PI | 4 | 6 | 7 | 7 | 4 | Over | 4 | 4 | 7 | Exp | 5 |
| 1 | C | MT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 3 | 5 | Exp | 4 |
| 1 | C | MT | 5 | 4 | 4 | 6 | 3 | Over | 3 | 3 | 6 | Pic | 3 |
| 1 | C | CT | 5 | 5 | 6 | 6 | 3 | Over | 3 | 2 | 5 | Pic | 3 |
| 1 | C | CT | 4 | 5 | 7 | 7 | 5 | Over | 4 | 4 | 6 | Exp | 4 |
| 3 | C | PI | 4 | 4 | 7 | 7 | 5 | Over | 5 | 4 | 6 | Exp | 4 |
| 3 | C | MT | 4 | 4 | 5 | 5 | 4 | Accurate | 3 | 3 | 6 | Pic | 5 |
| 3 | C | MT | 5 | 4 | 4 | 5 | 5 | Over | 4 | 3 | 5 | Exp | 3 |
| 3 | C | CT | 6 | 6 | 5 | 5 | 4 | Accurate | 3 | 5 | 6 | Exp | 6 |
| 3 | C | CT | 3 | 3 | 7 | 7 | 4 | Accurate | 3 | 3 | 6 | Exp | 5 |
| | | **Average** | **4.5** | **4.6** | **5.8** | **6.1** | **4.0** | | **3.5** | **3.4** | **5.8** | | **4.2** |
| 2 | C | PI | 3 | 3 | 7 | 5 | 5 | Over | 2 | 2 | 5 | Exp | 2 |
| 2 | C | MT | 5 | 5 | 3 | 5 | 4 | Over | 3 | 2 | 5 | Exp | 5 |
| 2 | C | MT | 5 | 4 | 3 | 3 | 4 | Over | 2 | 2 | 5 | Pic | 4 |
| 2 | C | CT | 2 | 2 | 5 | 6 | 4 | Over | 3 | 5 | 6 | Pic | 4 |
| 2 | C | CT | 7 | 7 | 1 | 3 | 5 | Over | 3 | 6 | 7 | Pic | 7 |
| 3 | H | PI | 4 | 3 | 5 | 6 | 5 | Over | 3 | 3 | 6 | Exp | 5 |
| 3 | H | MT | 5 | 4 | 5 | 5 | 4 | Accurate | 1 | 2 | 5 | Exp | 7 |
| 3 | H | MT | 3 | 2 | 6 | 6 | 3 | Under | 3 | 4 | 5 | Exp | 5 |
| 3 | H | CT | 7 | 6 | 6 | 3 | 6 | Over | 3 | 4 | 3 | Pic | 4 |
| 3 | H | CT | 3 | 3 | 7 | 7 | 7 | Over | 3 | 5 | 6 | Pic | 3 |
| | | **Average** | **4.4** | **3.9** | **4.8** | **4.9** | **4.7** | | **2.6** | **3.5** | **5.3** | | **4.6** |
| 1 | H | PI | 5 | 4 | 7 | 7 | 3 | Accurate | 4 | 3 | 7 | Pic | 1 |
| 1 | H | MT | 6 | 5 | 6 | 4 | 5 | Over | 1 | 4 | 6 | Exp | 4 |
| 1 | H | MT | 4 | 4 | 3 | 5 | 3 | Accurate | 4 | 3 | 6 | Pic | 2 |
| 1 | H | CT | 6 | 6 | 6 | 6 | 4 | Accurate | 6 | 5 | 7 | Pic | 7 |
| 1 | H | CT | 3 | 6 | 6 | 6 | 5 | Over | 3 | 6 | 6 | Exp | 6 |
| 2 | H | PI | 2 | 1 | 1 | 3 | 5 | Accurate | 3 | 2 | 3 | Pic | 1 |
| 2 | H | MT | 4 | 6 | 6 | 7 | 6 | Over | 2 | 3 | 5 | Exp | 4 |
| 2 | H | MT | 5 | 5 | 7 | 7 | 6 | Over | 4 | 4 | 5 | Exp | 6 |
| 2 | H | CT | 5 | 6 | 4 | 7 | 6 | Over | 4 | 4 | 6 | Pic | 6 |
| 2 | H | CT | 6 | 6 | 5 | 5 | 6 | Over | 2 | 4 | 7 | Pic | 6 |
| | | **Average** | **4.6** | **4.9** | **5.1** | **5.7** | **4.9** | | **3.3** | **3.8** | **5.8** | | **4.3** |